United States Patent	12406223
Kind Code	B2
Date of Patent	September 02, 2025
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Control device and method for controlling an unmanned delivery vehicle

Abstract

A control device (**100**) controls a delivery vehicle and includes a first determiner (**120**) that determines a point (i) by using each of a plurality of shipping locations as a reference and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at each of the plurality of shipping locations. Additionally, the control device (**100**) includes a controller (**130**) that allows the delivery vehicle to accept a shipping location of a package or a delivery destination of the package as a movement destination so as to set a state of the delivery vehicle to a state of accepting the movement destination, moves the delivery vehicle to the movement destination when the state of the delivery vehicle is the state of accepting the movement destination, and moves the delivery vehicle toward the point when the state of the delivery vehicle is a state of not accepting the movement destination.

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Appl. No.:	**17/795368**
Filed (or PCT Filed):	**May 31, 2021**
PCT No.:	**PCT/JP2021/020723**
PCT Pub. No.:	**WO2022/254539**
PCT Pub. Date:	December 08, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240185171 A1	Jun. 06, 2024

Publication Classification

Int. Cl.: **G06Q10/0834** (20230101)

U.S. Cl.:

CPC **G06Q10/0834** (20130101);

Field of Classification Search

CPC: G06Q (10/0834); G06Q (10/083)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9934530	12/2017	Iacono	N/A	G06Q 30/0639
10482421	12/2018	Ducrou	N/A	G06V 20/54
11610276	12/2022	Cohen	N/A	G06Q 10/06316
2020/0293989	12/2019	Son	N/A	N/A
2021/0241224	12/2020	Taniguchi	N/A	G06Q 50/40
2021/0248530	12/2020	Garcia-Brosa	N/A	G06Q 50/12
2022/0207478	12/2021	Hlavacek	N/A	G06N 5/01
2023/0359223	12/2022	Sakata	N/A	G05D 1/104

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020-7148	12/2019	JP	N/A
6629490	12/2019	JP	N/A
2020-149558	12/2019	JP	N/A
2020/013114	12/2019	WO	N/A

OTHER PUBLICATIONS

The IP.com Journal, (Dec. 10, 2020). Smart Variable Order Delivery Time or Order Pickup Time System. (Year: 2020). cited by examiner

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/JP2021/020723 filed May 31, 2021.

TECHNICAL FIELD

(2) The present disclosure relates to a control device and a method.

BACKGROUND ART

(3) In the related art, a system that notifies a terminal of a user, who is in a delivery destination, of the estimated time of arrival of a delivery vehicle, which is an unmanned delivery vehicle **80**, to the delivery destination, and then delivers a package, which is an ordered product **71***c*, to the delivery destination by the delivery vehicle is known (for example, Patent Literature 1). After the package is delivered, the system determines a vehicle waiting area **40** closest to the delivery vehicle as a point of an objective destination and moves the delivery vehicle to the determined point.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Unexamined Japanese Patent Application Publication No. 2020-007148.

SUMMARY OF INVENTION

Technical Problem

(5) However, the system of Patent Literature 1 determines the point of the objective destination not based on the time until shipment of the package becomes possible at a shipping location despite that start of the delivery of the package is not possible before the shipment of the package becomes possible. Therefore, the system of Patent Literature 1 has a first problem in that it is not possible to suppress the difference between the timing when the delivery vehicle arrives at the shipping location from the point of the objective destination or a point on the way to the point of the objective destination and the timing when shipment of the package becomes possible at the shipping location.

(6) Furthermore, the system of Patent Literature 1 has a second problem in that since a mere notification to the delivery destination is performed, it is not possible to notify the shipping location of the start timing of preparation in order to suppress the difference between the timing when the delivery vehicle arrives at the shipping location and the timing when preparation of the package is completed and shipment of the package becomes possible at the shipping location.

(7) The present disclosure has been made to solve the above problems, and a first objective of the present disclosure is to provide a control device and a method, capable of suppressing the difference between the timing when a delivery vehicle arrives at a shipping location and the timing when preparation of a package is completed and shipment of the package becomes possible at the shipping location.

(8) A second objective of the present disclosure is to provide a control device and a method, capable of notifying a shipping location of the start timing of preparation in order to suppress the difference between the timing when a delivery vehicle arrives at the shipping location and the timing when preparation of a package is completed and shipment of the package becomes possible at the shipping location.

Solution to Problem

(9) In order to achieve the above first objective, a control device according to a first aspect of the present disclosure is a control device for controlling a delivery vehicle, the control device including: a determiner that determines a point (i) by using each of a plurality of shipping locations as a reference and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at each of the plurality of shipping locations; and a

controller that allows the delivery vehicle to accept a shipping location of a package or a delivery destination of the package as a movement destination so as to set a state of the delivery vehicle to a state of accepting the movement destination, moves the delivery vehicle to the movement destination when the state of the delivery vehicle is the state of accepting the movement destination, and moves the delivery vehicle toward the point when the state of the delivery vehicle is a state of not accepting the movement destination.
(10) In order to achieve the above second objective, a control device according to a second aspect of the present disclosure is a control device for controlling a delivery vehicle, the control device including: a communicator that transmits, to a terminal device at a shipping location, a notification notifying that a start timing of preparation has arrived when the delivery vehicle arrives at a boundary of a region determined (i) by using, as a reference, the shipping location accepted as a movement destination and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at the shipping location.
Advantageous Effects of Invention
(11) In accordance with a control device and a method according to a first aspect of the present disclosure, it is possible to suppress the difference between the timing when a delivery vehicle arrives at a shipping location and the timing when preparation of a package is completed and shipment of the package becomes possible at the shipping location.
(12) In accordance with a control device and a method according to a second aspect of the present disclosure, it is possible to notify a shipping location of the start timing of preparation in order to suppress the difference between the timing when a delivery vehicle arrives at the shipping location and the timing when preparation of a package is completed and shipment of the package becomes possible at the shipping location.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a system configuration diagram illustrating a configuration example of a delivery system according to the present disclosure;
(2) FIG. **2** is a diagram illustrating an example of a point of an objective destination located outside a neighboring region;
(3) FIG. **3** is a hardware configuration diagram illustrating a configuration example of a control device;
(4) FIG. **4** is a flowchart illustrating an example of an objective destination determination process performed by the control device;
(5) FIG. **5** is a functional block diagram illustrating an example of a function of the control device;
(6) FIG. **6** is a diagram illustrating an example of an estimated preparation time table stored in the control device;
(7) FIG. **7** is a front part of a flowchart showing an example of a movement control process performed by the control device;
(8) FIG. **8** is a central part of the flowchart showing the example of the movement control process performed by the control device;
(9) FIG. **9** is a rear part of the flowchart showing the example of the movement control process performed by the control device;
(10) FIG. **10** is a diagram illustrating an example of a task table stored in the control device;
(11) FIG. **11** is an external appearance configuration diagram illustrating an external appearance example of a delivery vehicle according to the embodiment;
(12) FIG. **12** is a hardware configuration diagram illustrating a configuration example of a control device included in the delivery vehicle;

(13) FIG. **13** is a flowchart illustrating an example of an objective destination movement process performed by the delivery vehicle;
(14) FIG. **14** is a flowchart illustrating an example of a task generation process performed by the control device;
(15) FIG. **15** is a flowchart illustrating an example of a movement destination movement process performed by the delivery vehicle;
(16) FIG. **16** is a flowchart illustrating an example of a notification control process performed by the control device;
(17) FIG. **17** is a diagram illustrating an example of a preparation time table stored in the control device;
(18) FIG. **18** is a diagram illustrating an example of a point of an objective destination located inside a neighboring region;
(19) FIG. **19** is a diagram illustrating an example of a point of an objective destination located at a point of tangency between two neighboring regions;
(20) FIG. **20** is a diagram illustrating an example of a point of an objective destination located at a point of intersection of two boundaries of neighboring regions; and
(21) FIG. **21** is an external appearance configuration diagram illustrating an external appearance example of a delivery vehicle according to modification 22 of the embodiment.

DESCRIPTION OF EMBODIMENTS

Embodiments

(22) Hereinafter, embodiments of the present disclosure are described with reference to the accompanying drawings.
(23) A delivery system **1** according to an embodiment of the present disclosure includes a control device **100** as illustrated in FIG. **1** that controls the delivery of an ordered package, and a delivery vehicle **600** that stores the ordered package and moves under the control of the control device **100** in order to deliver the ordered package to a delivery destination. In the present embodiment, the following description is given by taking, as a specific example, a case where a package is an ordered product. Furthermore, the following description is given on the assumption that an order for a product includes a shipping request for requesting the shipment of the product, a collection request for collecting the shipped product, and a delivery request for requesting delivery of the shipped and collected product to a delivery destination.
(24) Furthermore, the delivery system **1** includes a terminal device **800** carried by an orderer who orders a product. Moreover, the delivery system **1** includes a terminal device **901** carried by an employee of a store ST**1** that sells ordered products and a terminal device **902** carried by an employee of a store ST**2** that sells ordered products.
(25) The store ST**1** has an entrance that is a shipping location R**1** as illustrated in FIG. **2** where products are shipped. The store ST**2** has an entrance that is a shipping location R**2**.
(26) The control device **100** is a server device, and is installed in, for example, an office building of an intermediary who mediates the sale of products. The control device **100** includes a central processing unit (CPU) **101**, a random access memory (RAM) **102**, a read only memory (ROM) **103***a*, a hard disk **103***b*, a data communication circuit **104**, a video card **105***a*, a display device **105***b*, and an input device **105***c* as illustrated in FIG. **3**, which are hardware. The control device **100** may include a plurality of CPUs, or a plurality of RAMs and flash memories.
(27) The CPU **101** of the control device **100** performs the overall control of the entire control device **100** by executing a program stored in the ROM **103***a* or the hard disk **103***b*. The RAM **102** temporarily stores data to be processed when the program is executed by the CPU **101**. The ROM **103***a* and the hard disk **103***b* store various programs, various data used for executing the programs, and a table in which the data are stored.
(28) The data communication circuit **104** of the control device **100** is a network interface card (NIC), and, for example, performs data communication with a not-illustrated base station

connected to the Internet IN by using radio waves according to communication standards such as long term evolution (LTE) and 5th generation (5G). With this, the data communication circuit **104** of the control device **100** performs data communication with the terminal devices **800**, **901**, and **902** and the delivery vehicle **600** that are connected to the Internet IN.
(29) The video card **105***a* of the control device **100** renders an image on the basis of a digital signal output from the CPU **101**, and outputs an image signal representing the rendered image. The display device **105***b* is an electroluminescence (EL) display, a plasma display panel (PDP), or a liquid crystal display (LCD), and displays the image according to the image signal output from the video card **105***a*. The input device **105***c* is one or more of a keyboard, a mouse, a touch pad, and a button, and inputs a signal corresponding to an operation of an employee of the intermediary.
(30) When the control device **100** is started, the CPU **101** of the control device **100** determines a point PD of an objective destination as illustrated in FIG. **2** by performing an objective destination determination process as illustrated in FIG. **4**. With this, the CPU **101** of the control device **100** serves as an acquirer **110** as illustrated in FIG. **5** that acquires estimated preparation time information representing an estimated preparation time for each of the two shipping locations R**1** and R**2** that can be specified in the order of a product. On the assumption that a shipping request for a product estimated to be ordered (hereinafter, referred to as an estimated product) has been received, the estimated preparation time is the preparation time required from when the shipping request is accepted until when preparation of the estimated product is completed and shipment of the estimated product becomes possible.
(31) In the present embodiment, the fact that shipment of the product becomes possible means that the state of the product changes to a state in which receiving of the product by the delivery vehicle **600** is possible. Furthermore, in the present embodiment, the state in which receiving of the product by the delivery vehicle **600** is possible includes a state in which the delivery of the product by the delivery vehicle **600** is startable.
(32) In the present embodiment, the product is a food and drink. Therefore, the state of the product of which delivery is startable includes a state of having been cooked if the product needs to be cooked by the start of delivery of the product. Furthermore, the state of the product of which delivery is startable includes a state of having been packaged if the product needs to be packaged by the start of delivery of the product. Whether or not the product needs to be cooked or needs to be packaged is explicitly or implicitly agreed between an orderer and the employee of the store ST**1** or ST**2** at the time of accepting the order for the product, or is predetermined by business practice. Furthermore, the state of the product of which delivery is startable includes a state in which the product is located at the shipping location R**1** or R**2** of the store ST**1** or ST**2** specified in the order of the product.
(33) Therefore, when a product does not need to be cooked and packaged, preparations for making shipment of the product possible include, for example, searching for the product from the shelves of a warehouse and transporting the searched product to the shipping location R**1** or R**2** of the product. Therefore, the preparation time includes the search time required for finding the product and the transportation time required for transporting the found product to the shipping location R**1** or R**2**.
(34) Furthermore, when a product does not need to be cooked but needs to be packaged, the preparation of the product includes finding the product, packaging the product, and transporting the product to the shipping location R**1** or R**2**. Therefore, the preparation time of the product is the sum of the search time of the product, the packaging time required for packaging the found product, and the transportation time of the packaged product to the shipping location R**1** or R**2**.
(35) Moreover, when a product needs to be cooked and packaged, the preparation of the product includes cooking of the product, packaging of the cooked product, and transportation of the packaged product to the shipping location R**1** or R**2**. Therefore, the preparation time of the product includes the sum of the cooking time required for cooking the product, the packaging time of the

cooked product, and the transportation time of the packaged product to the shipping location R**1** or R**2**.
(36) The CPU **101** of the control device **100** serves as a first determiner **120** that determines the point PD of the objective destination of the delivery vehicle **600** on the basis of the estimated preparation time information acquired for each of the shipping locations R**1** and R**2**. Furthermore, the CPU **101** serves as a controller **130** that performs a control for moving the delivery vehicle **600** toward the determined point PD.
(37) The hard disk **103***b* of the control device **100** serves as an information storage **190** that stores information used for performing the objective destination determination process. The information storage **190** stores in advance an estimated preparation time table as illustrated in FIG. **6**, in which the estimated preparation time information is stored in advance.
(38) A plurality of records is stored in advance in the estimated preparation time table. In each record of the estimated preparation time table, a store ID (IDentification) identifying the store ST**1** or ST**2** and the estimated preparation time information representing the estimated preparation time of the store ST**1** or ST**2** are stored in advance in correlation with each other.
(39) In the present embodiment, an estimated product in the store ST**1** is a product with the highest probability that an order is confirmed among a plurality of products that is sold in the store ST**1**. The probability that an order for a product sold in store ST**1** is confirmed is a value obtained by dividing the number of previously confirmed orders specifying the shipping location R**1** in the store ST**1** and specifying the product by the number of previously confirmed orders specifying the shipping location R**1** in the store ST**1**. Similarly, an estimated product in the store ST**2** is a product with the highest probability that an order is confirmed among a plurality of products that is sold in the store ST**2**.
(40) When the execution of the objective destination determination process of FIG. **4** is started, the acquirer **110** of the control device **100** acquires estimated preparation time information from the estimated preparation time table of FIG. **6** for each of the two shipping locations R**1** and R**2** (step S**01**). For this purpose, the first determiner **120** determines numbers of the shipping locations R**1** and R**2** according to a predetermined rule or a software random number. In the present embodiment, the following description is given by taking, as an example, a case where the number of the shipping location R**1** is determined to be No. 1 and the number of the shipping location R**2** is determined to be No. 2.
(41) Next, the acquirer **110** of the control device **100** acquires, from the estimated preparation time table, estimated preparation time information (hereinafter, referred to as first estimated preparation time information) correlated with the store ID of the store ST**1** including the first shipping location R**1**. Similarly, the acquirer **110** acquires, from the estimated preparation time table, estimated preparation time information (hereinafter, referred to as second estimated preparation time information) correlated with the store ID of the store ST**2** including the second shipping location R**2**.
(42) Next, the acquirer **110** of the control device **100** acquires position information stored in advance in the information storage **190** with respect to the store ID of the store ST**1** including the first shipping location R**1**, thereby acquiring position information representing the position of the shipping location R**1** using latitude, longitude, and altitude. Next, for the case of k=1 and the case of k=2, the first determiner **120** determines a neighboring region (hereinafter, referred to as a k-th neighboring region) where the time required for the delivery vehicle **600** to move to an k-th shipping location is equal to or less than an estimated preparation time represented by k-th estimated preparation time information.
(43) In the present embodiment, the delivery vehicle **600** moves at a preset setting speed. The acquirer **110** of the control device **100** acquires information representing the setting speed of the delivery vehicle **600** stored in advance in the information storage **190**. Thereafter, the first determiner **120** multiplies the setting speed of the delivery vehicle **600** represented by the acquired

information by an estimated preparation time represented by the first estimated preparation time information, thereby calculating a distance D**1** in which the delivery vehicle **600** moves in the estimated preparation time.
(44) Next, the first determiner **120** of the control device **100** determines a boundary B**1** as illustrated in FIG. **2**, which is separated by the calculated distance D**1** from the position of the shipping location R**1** represented by the acquired position information. Thereafter, the first determiner **120** determines the determined boundary B**1** and a region on the shipping location R**1** side from the boundary B**1** as a first neighboring region N**1**.
(45) Thereafter, similarly, the first determiner **120** of the control device **100** determines a neighboring region (hereinafter, referred to as a second neighboring region) N**2** where the time required for the delivery vehicle **600** to move to the second shipping location R**2** is equal to or less than an estimated preparation time represented by the second estimated preparation time information (step S**02**).
(46) For this purpose, the first determiner **120** of the control device **100** calculates a distance D**2** in which the delivery vehicle **600** moves in the estimated preparation time represented by the second estimated preparation time information, and determines a boundary B**2** separated from the position of the shipping location R**2** by the calculated distance D**2**. Thereafter, the first determiner **120** determines the boundary B**2** and a region on the shipping location R**2** side from the boundary B**2** as a second neighboring region N**2**.
(47) In the present embodiment, since the distances D**1** and D**2** are Euclidean distances in a three-dimensional space, the neighboring regions N**1** and N**2** are spherical regions, and the boundaries B**1** and B**2** of the neighboring regions N**1** and N**2** are spherical surfaces, respectively.
(48) The controller **130** of the control device **100** stores neighboring region information representing the first neighboring region N**1** in the information storage **190** in correlation with the store ID of the store ST**1** including the first shipping location R**1**. Similarly, the controller **130** stores neighboring region information representing the second neighboring region N**2** in the information storage **190** in correlation with the store ID of the store ST**2** including the second shipping location R**2** (step S**03**). In the present embodiment, the neighboring region information representing the first neighboring region N**1** is information representing an equation that defines the boundary B**1** of the first neighboring region N**1**, and the neighboring region information representing the second neighboring region N**2** is information representing an equation that defines the boundary B**2** of the second neighboring region N**2**; however, the present disclosure is not limited thereto.
(49) Thereafter, the first determiner **120** of the control device **100** determines the point PD so that region distances L**1** and L**2**, which are distances from the point PD of the objective destination to the two boundaries B**1** and B**2**, satisfy a predetermined condition (step S**04**). In the present embodiment, the region distances L**1** and L**2** are Euclidean distances in the three-dimensional space.
(50) In the present embodiment, the predetermined condition is a condition that the sum of the region distance L**1** from the point PD of the objective destination to the boundary B**1** and the region distance L**2** from the point PD to the boundary B**2** is minimum. Therefore, the first determiner **120** determines the point PD of the objective destination, where the sum of the region distance L**1** and the region distance L**2** is minimized, by using, for example, a known algorithm including a least squares method.
(51) Thereafter, the controller **130** of the control device **100** stores the position information representing, using latitude, longitude, and altitude, the position of the determined point PD of the objective destination in the information storage **190** (step S**05**), and then ends the execution of the objective destination determination process.
(52) When the execution of the objective destination determination process is ended, the CPU **101** of the control device **100** performs a movement control process as illustrated in FIG. **7** to FIG. **9**, in

order to control the movement of the delivery vehicle **600**. The information storage **190** stores in advance a task table as illustrated in FIG. **10**, which is used for performing the movement control process. To the task table, records are to be added, in each of which a task ID identifying a collection delivery task for a product of which an order has been confirmed, a user ID of an orderer who placed the order, a product ID of the product, position information of the shipping location R**1** or R**2** specified in the order of the product, and position information of a delivery destination of the product are stored in correlation with one another. In the present embodiment, the confirmation of an order means that all of a shipping request, a collection request, and a delivery request included in the order are accepted.

(53) The collection delivery task includes a collection task and a delivery task. The collection task for a product is a task of allowing the delivery vehicle **600** to collect the product at the shipping location R**1** or R**2** specified in the order of the product. Furthermore, the delivery task for a product is a task of allowing the delivery vehicle **600** to deliver the collected product to a delivery destination of the product. In the present embodiment, since the collection delivery task has not been generated at the start of execution of the movement control process, the following description is given on the assumption that no record is stored in the task table.

(54) When the execution of the movement control process of FIG. **7** is started, the controller **130** of the control device **100** determines that the state of the delivery vehicle **600** is a state (hereinafter, referred to as a non-accepting state) different from a state (hereinafter, referred to as an accepting state) in order to execute the collection task or the delivery task, the accepting state being a state of accepting, by the delivery vehicle **600**, the shipping location R**1** or R**2** or the delivery destination as a movement destination (step S**11**). The non-accepting state is a state of not accepting, by the delivery vehicle **600**, the shipping locations R**1** or R**2** or the delivery destination as the movement destination. Thereafter, the controller **130** initializes the value of a state flag with a value indicating the non-accepting state, the value of the state flag being stored in advance in the information storage **190** and indicating whether or not the state of the delivery vehicle **600** is the accepting state or the non-accepting state.

(55) Next, the acquirer **110** of the control device **100** attempts to acquire the task ID from the task table of FIG. **10** (step S**12**). Next, the acquirer **110** determines whether or not the task ID has been acquired (step S**13**). When it is determined that the task ID has not been acquired (step S**13**; No), the controller **130** determines that the collection delivery task does not exist, that is, the collection task and the delivery task do not exist.

(56) Next, the acquirer **110** of the control device **100** outputs a transmission request for requesting transmission of the position information representing the position of the delivery vehicle **600** using the latitude, longitude, and altitude to the data communication circuit **104** with the delivery vehicle **600** as a destination. When the data communication circuit **104** of the control device **100** transmits the transmission request to the delivery vehicle **600** and then receives the position information from the delivery vehicle **600**, the acquirer **110** acquires the position information of the delivery vehicle **600** from the data communication circuit **104**. Furthermore, the acquirer **110** acquires the position information of the point PD of the objective destination stored in step S**05** of FIG. **4** from the information storage **190** (step S**14**).

(57) Thereafter, the controller **130** of the control device **100** determines whether or not the position of the delivery vehicle **600** represented by the acquired position information matches the position of the point PD of the objective destination represented by the position information (step S**15**). When it is determined that the position of the delivery vehicle **600** matches the position of the point PD of the objective destination (step S**15**; Yes), the controller **130** determines that the delivery vehicle **600** is located at the point PD of the objective destination. Therefore, the controller **130** repeats the above process from step S**12** without performing the process of moving the delivery vehicle **600** toward the point PD.

(58) However, when it is determined that the position of the delivery vehicle **600** is different from

the position of the point PD of the objective destination (step S**15**; No), the controller **130** of the control device **100** determines that the delivery vehicle **600** is located at a point different from the point PD of the objective destination. Next, the controller **130** generates a movement command including the position information of the point PD of the objective destination and commanding the movement of the delivery vehicle **600** toward the point PD, and outputs the generated movement command to the data communication circuit **104** with the delivery vehicle **600** as a destination (step S**16**). With this, the controller **130** perform a control for moving the delivery vehicle **600** toward the point PD of the objective destination.

(59) Next, the acquirer **110** of the control device **100** attempts to acquire an arrival report from the data communication circuit **104**. The arrival report is a report received from the delivery vehicle **600** and is a report notifying that the delivery vehicle **600** is arrived at the point PD of the objective destination. When the acquirer **110** determines that the arrival report has been acquired (step S**17**; Yes), the acquirer **110** repeats the above process from step S**12**.

(60) However, when it is determined that the arrival report has not been acquired (step S**17**; No), the acquirer **110** of the control device **100** attempts to acquire a task ID by performing the same process as that of step S**12** (step S**18**). The acquisition of the task ID is attempted because the collection delivery task may be generated while the delivery vehicle **600** is heading for the point PD of the objective destination. When the acquirer **110** determines that the task ID has not been acquired (step S**19**; No), the acquirer **110** of the control device **100** repeats the above process from step S**17**.

(61) Before describing a process to be performed after it is determined that the task ID has been acquired in step S**13** or S**19**, the description of the movement control process is interrupted and the delivery vehicle **600** and the like are described.

(62) The delivery vehicle **600** is an unmanned ground vehicle as illustrated in FIG. **11**. The delivery vehicle **600** includes a chassis **610** having a plurality of wheels including wheels **601** and **602**, a storage cabinet **620** installed on an upper surface of the chassis **610**, and a control device **690** built in the storage cabinet **620**.

(63) The storage cabinet **620** of the delivery vehicle **600** includes one storage box **621** capable of storing products. The storage box **621** includes a box body that includes a not-illustrated bottom plate, top plate, and back plate, and two not-illustrated side plates, and that is closed by these plates to form a space where the front is open. A door frame **621***b* for receiving a door **621***a* is installed at the opening of the box body.

(64) The door **621***a* is provided on the back surface thereof with a dead bolt **621***c* that is a bolt, and the door frame **621***b* is provided with a strike **621***d* that is a receiving seat for the dead bolt **621***c*. The door **621***a* further includes a not-illustrated motor that locks the door **621***a* by inserting the dead bolt **621***c* into the strike **621***d* according to a signal output from the control device **690**. The motor unlocks the door **621***a* by pulling out the dead bolt **621***c* from the strike **621***d* according to a signal output from the control device **690**.

(65) The storage cabinet **620** of the delivery vehicle **600** includes an imaging device **630**, which is a digital camera, for example, on the surface installed with the door **621***a*. An optical axis and a view angle of the imaging device **630** are adjusted so that a recipient who receives products stored in the storage box **621** is included in an imaging range. The imaging device **630** performs imaging at a predetermined cycle, and outputs a signal representing an image obtained by the imaging to the control device **690**.

(66) The delivery vehicle **600** further includes a light detection and ranging (LiDAR) sensor **641** installed on a front surface of the chassis **610** and a not-illustrated LiDAR sensor installed on a rear surface of the chassis **610**.

(67) When the front direction of the delivery vehicle **600** is used as a reference azimuth, the LiDAR sensor **641** on the front surface included in the delivery vehicle **600** emits laser light in a plurality of directions included in a range in which an azimuth angle formed with the reference azimuth is

−90 degrees to +90 degrees and an elevation angle formed with the front direction of the delivery vehicle **600** is −90 degrees to +90 degrees. The LiDAR sensor **641** on the front surface receives reflected light of the emitted laser light, and measures distances to a plurality of reflection points, where the laser light is reflected, on the basis of the time from the emission of the laser light to the reception of the reflected light. Next, the LiDAR sensor **641** on the front surface calculates coordinate values of the plurality of reflection points in a three-dimensional coordinate system using a center point of the delivery vehicle **600** as the origin, on the basis of the emission direction of the laser light and the measured distances. Thereafter, the LiDAR sensor **641** on the front surface outputs the calculated coordinate values of the plurality of reflection points to the control device **690** of the delivery vehicle **600**.

(68) When the rear direction of the delivery vehicle **600** is used as a reference azimuth, the LiDAR sensor on the rear surface included in the delivery vehicle **600** emits infrared laser light in a plurality of directions included in a range in which an azimuth angle formed with the reference azimuth is −90 degrees to +90 degrees and an elevation angle formed with the rear direction of the delivery vehicle **600** is −90 degrees to +90 degrees. Furthermore, the LiDAR sensor on the rear surface calculates coordinate values of a plurality of reflection points of the emitted laser light in the three-dimensional coordinate system of the delivery vehicle **600**, and outputs the calculated coordinate values of the plurality of reflection points to the control device **690** of the delivery vehicle **600**.

(69) The output of the coordinate values of the plurality of reflection points from the LiDAR sensor **641** on the front surface and the LiDAR sensor on the rear surface included in the delivery vehicle **600** to the control device **690** is for the control device **690** of the delivery vehicle **600** to specify coordinate values, sizes, and the like in the three-dimensional space of an object in all directions using the delivery vehicle **600** as a reference, in order to travel by avoiding the object such as an obstacle.

(70) The control device **690** of the delivery vehicle **600** includes a CPU **691**, a RAM **692**, a ROM **693***a*, a flash memory **693***b*, a data communication circuit **694**, a video card **695***a*, a display device **695***b*, an input device **695***c*, a position measurement circuit **696**, an input/output port **698**, and a driving circuit **699**, which are hardware as illustrated in FIG. **12**. The delivery vehicle **600** may include a plurality of CPUs, or may include a plurality of RAMs and flash memories.

(71) The configurations and functions of the CPU **691**, the RAM **692**, the ROM **693***a*, the data communication circuit **694**, the video card **695***a*, the display device **695***b*, and the input device **695***c* of the delivery vehicle **600** are the same as those of the CPU **101**, the RAM **102**, the ROM **103***a*, the data communication circuit **104**, the video card **105***a*, the display device **105***b*, and the input device **105***c* of the control device **100**.

(72) The flash memory **693***b* of the delivery vehicle **600** stores various programs, various data used for executing the programs, and a table in which the data are stored. The delivery vehicle **600** may include a hard disk instead of the flash memory **693***b*.

(73) The position measurement circuit **696** of the delivery vehicle **600** is a quasi-zenith satellite system (QZSS) circuit. The position measurement circuit **696** receives signals output from quasi-zenith satellites, measures latitude, longitude, and altitude indicating the position of the delivery vehicle **600** on the basis of the received signals, and outputs a signal representing the measured latitude, longitude, and altitude. The position measurement circuit **696** may not be the QZSS circuit, but may be a global positioning system (GPS) circuit that receives GPS signals output from GPS satellites, and measures latitude, longitude, and altitude representing the position of the delivery vehicle **600** on the basis of the received GPS signals.

(74) The input/output port **698** of the delivery vehicle **600** is connected to a not-illustrated cable connected to the imaging device **630**, and inputs a signal output by the imaging device **630** to the CPU **691**. Furthermore, the input/output port **698** is connected to not-illustrated cables connected to the LiDAR sensor **641** on the front surface and the LiDAR sensor on the rear surface, respectively.

The input/output port **698** inputs, to the CPU **691**, a signal representing the coordinate values output by each of the LiDAR sensor **641** on the front surface and the LiDAR sensor on the rear surface.

(75) The driving circuit **699** of the delivery vehicle **600** is connected to not-illustrated cables respectively connected to not-illustrated motors for rotating the plurality of wheels. The driving circuit **699** rotates the plurality of wheels by driving the motors according to a control signal output by the CPU **691**.

(76) Furthermore, the driving circuit **699** of the delivery vehicle **600** is connected to not-illustrated cables respectively connected to the not-illustrated motor for pulling out the dead bolt **621***c* included in the door **621***a* from the strike **621***d* or inserting the dead bolt **621***c* into the strike **621***d*, and drives the motor according to a control signal output by the CPU **691**.

(77) When the data communication circuit **694** of the delivery vehicle **600** receives the transmission request output in step S**14** of FIG. **7** from the control device **100**, the CPU **691** of the delivery vehicle **600** specifies the latitude, longitude, and altitude of the delivery vehicle **600** on the basis of a signal output from the position measurement circuit **696**. Next, the CPU **691** generates position information representing the position of the delivery vehicle **600** using the latitude, longitude, and altitude, and outputs the generated position information to the data communication circuit **694** with the control device **100** as a destination. Thereafter, the data communication circuit **694** of the delivery vehicle **600** transmits the position information of the delivery vehicle **600** to the control device **100**.

(78) When the data communication circuit **694** of the delivery vehicle **600** receives the movement command output in step S**16** of FIG. **7**, the CPU **691** of the delivery vehicle **600** performs an objective destination movement process as illustrated in FIG. **13** in order to move according to the movement command.

(79) When the execution of the objective destination movement process is started, the CPU **691** of the delivery vehicle **600** acquires the movement command from the data communication circuit **694**, and acquires the position information of the point PD of the objective destination from the acquired movement command (step S**51**). Next, the CPU **691** specifies the latitude, longitude, and altitude of the delivery vehicle **600** on the basis of a signal output from the position measurement circuit **696**. Thereafter, the CPU **691** determines a movement route from the specified position of the delivery vehicle **600** to the point PD of the objective destination.

(80) For this purpose, the CPU **691** of the delivery vehicle **600** reads a plurality of records from a partial route table stored in advance in the flash memory **693***b*. In each of the plurality of records in the partial route table, information relating to a partial route, such as, for example, a road and a sidewalk, along which the delivery vehicle **600** can move, is stored. The information relating to a partial route is information in which the latitude, longitude, and altitude of a start node of an edge, which is the partial route, the latitude, longitude, and altitude of an end node of the edge, and information indicating the distance of the edge are correlated with one another.

(81) Next, the CPU **691** of the delivery vehicle **600** executes a known route search algorithm including, for example, a Dijkstra method, by using the distance of the edge, the latitude, longitude, and altitude of the start node and the end node of the edge, which are stored in the respective read records of the partial route table, the latitude, longitude, and altitude representing the position of the delivery vehicle **600**, and the latitude, longitude, and altitude representing the point PD of the objective destination. With this, the CPU **691** searches for the shortest overall route from the position of the delivery vehicle **600** to the point PD of the objective destination by combining the partial routes, and determines the searched overall route as a movement route of the delivery vehicle **600** (step S**52**). Thereafter, the CPU **691** of the delivery vehicle **600** generates information representing a movement route including information representing, using latitude, longitude, and altitude, the positions of nodes included in the determined movement route, and information representing the passage sequence of the nodes.

(82) Next, the CPU **691** of the delivery vehicle **600** acquires information representing the setting speed of the delivery vehicle **600** stored in advance in the flash memory **693***b*. Thereafter, the CPU **691** specifies the latitude, longitude, and altitude of the delivery vehicle **600** on the basis of a signal output from the position measurement circuit **696**. Next, the CPU **691** generates a control signal for moving at the setting speed so as to reduce the difference between the specified latitude, longitude, and altitude and the latitude, longitude, and altitude of a node having the earliest passage order among a plurality of unpassed nodes included in the movement route, and outputs the generated control signal to the driving circuit **699** (step S**53**). With this, the delivery vehicle **600** moves at the setting speed toward the unpassed node having the earliest passing order.
(83) Thereafter, the CPU **691** of the delivery vehicle **600** attempts to acquire an acceptance command from the data communication circuit **694**. The acceptance command is a command received from the control device **100**, and is a command that commands accepting the shipping location R**1** or R**2** or the delivery destination as a movement destination. In the present embodiment, when an order is confirmed during a period from when the reception of the movement command until when the arrival at the point PD of the objective destination, the acceptance command is transmitted from the control device **100** to the delivery vehicle **600** in order to allow the delivery vehicle **600** to accept the shipping location R**1** or R**2** specified in the confirmed order as a movement destination.
(84) When the CPU **691** of the delivery vehicle **600** determines that the acceptance command has been acquired (step S**54**; Yes), the CPU **691** ends the execution of the objective destination movement process. With this, the delivery vehicle **600** stops movement toward the point PD of the objective destination. Thereafter, the delivery vehicle **600** moves to the shipping location R**1** or R**2**, which is a movement destination, or the delivery destination under the control of the control device **100**.
(85) However, when it is determined that the acceptance command has not been acquired (step S**54**; No), the CPU **691** of the delivery vehicle **600** determines whether or not the delivery vehicle **600** has arrived at the point PD of the objective destination, which is the end point of the movement route, on the basis of whether or not the delivery vehicle **600** has arrived at all of the plurality of nodes included in the movement route (step S**55**). When it is determined that the delivery vehicle **600** has not arrived at the end point of the movement route (step S**55**; No) because the delivery vehicle **600** has not arrived at all of the plurality of nodes, the CPU **691** repeats the above process from step S**53**.
(86) However, when it is determined that the delivery vehicle **600** has arrived at the end point of the movement route (step S**55**; Yes) because the delivery vehicle **600** has arrived at all of the plurality of nodes, the CPU **691** outputs a control signal for parking or stopping the delivery vehicle **600** at the position of the end point to the driving circuit **699**. Thereafter, the CPU **691** outputs an arrival report to the data communication circuit **694** with the control device **100** as a destination (step S**56**), and ends the execution of the objective destination movement process, the arrival report notifying that the delivery vehicle **600** is arrived at the point PD of the objective destination that is the end point of the movement route.
(87) When the data communication circuit **104** of the control device **100** receives an order request from the terminal device **800** of the orderer, the CPU **101** of the control device **100** performs a task generation process as illustrated in FIG. **14**, in order to generate a collection task and a delivery task for an ordered product. The task generation process may be performed before the movement command is transmitted and then the delivery vehicle **600** arrives at the point PD of the objective destination according to the movement command, may be performed when the delivery vehicle **600** has arrived at the point PD of the objective destination, or may be performed after the delivery vehicle **600** arrives at the point PD of the objective destination.
(88) In the present embodiment, the following description is given by taking, as a specific example, a case where the store ST**1** and a not-illustrated product I**1** sold in the store ST**1** are specified in an

order requested to be accepted by an order request. Furthermore, in the present embodiment, the following description is given on the assumption that specifying the store ST**1** means that the shipping location R**1** in the store ST**1** is specified.
(89) By performing the task generation process, the CPU **101** of the control device **100** further serves as a second determiner **140** as illustrated in FIG. **5** for determining whether or not a shipping request for the ordered product I**1** has been accepted in the store ST**1**.
(90) When the execution of the task generation process is started, the acquirer **110** of the control device **100** acquires the order request received from the terminal device **800** of the orderer from the data communication circuit **104**. Next, the acquirer **110** acquires a user ID of the orderer, a store ID identifying the specified store ST**1**, a product ID identifying the specified product I**1**, and information representing the delivery destination of the product I**1** by an address. Next, the acquirer **110** acquires information representing latitude, longitude, and altitude stored in advance in the information storage **190** in correlation with information representing the address of the delivery destination, thereby acquiring position information representing the position of the delivery destination using latitude, longitude, and altitude (step S**61**).
(91) Next, the controller **130** of the control device **100** outputs the order request to the data communication circuit **104** with the terminal device **901** of the store ST**1** identified by the acquired store ID as a destination. With this, the order request is transferred to the store ST**1** (step S**62**).
(92) When the data communication circuit **104** of the control device **100** does not receive an acceptance notification after when the order request is transferred and before when a predetermined time elapses from the transferring, the second determiner **140** determines that no order has been confirmed (step S**63**; No). In such a case, the reason why it is determined that no order has been confirmed is because the second determiner **140** determines that a shipping request has not been accepted among the shipping request, a collection request, and a delivery request included in the order requested to be accepted by the order request. Thereafter, the controller **130** outputs anon-acceptance notification notifying that the order has not been accepted to the data communication circuit **104** with the terminal device **800** of the orderer as a destination, and then ends the execution of the task generation process.
(93) However, when the acceptance notification is received after when the order request is transferred and before when the predetermined time elapses from the transferring, the second determiner **140** of the control device **100** determines that the shipping request has been accepted. Next, the second determiner **140** determines that the collection request and the delivery request of the product to be shipped are accepted, and determines that the order has been confirmed (step S**63**; Yes).
(94) Next, the acquirer **110** of the control device **100** acquires the position information of the shipping location R**1** stored in advance in the information storage **190** in correlation with the store ID of the store ST**1** acquired in step S**61** (step S**64**). The store ID of the store ST**1** and the position information of the shipping location R**1** are correlated with each other because the entrance of the store ST**1** is the shipping location R**1**.
(95) Thereafter, the controller **130** of the control device **100** generates a task ID on the basis of a predetermined rule or a software random number, and the acquirer **110** acquires a system date and time from, for example, an operation system (OS) as a task generation date and time. Next, the controller **130** stores information indicating the task generation date and time, the generated task ID, the user ID and the product ID acquired in step S**61**, the acquired position information of the shipping location R**1**, and the acquired position information of the delivery destination in the task table of FIG. **10** in correlation with one another. In this way, the controller **130** generates a collection delivery task including a collection task of allowing the delivery vehicle **600** to collect the product I**1** at the shipping location R**1** specified by the orderer, and a delivery task of allowing the delivery vehicle **600** to deliver the collected product I**1** to the delivery destination specified by the orderer (step S**65**). Thereafter, the controller **130** ends the execution of the task generation

process.
(96) Next, the interrupted movement control process is described again. When it is determined in step S**13** or S**19** of FIG. **7** that the acquirer **110** of the control device **100** has acquired the task ID of the task generated in step S**65** of FIG. **14** (step S**13**; Yes or step S**19**; Yes), it is determined that the collection delivery task exists, that is, at least one of the collection task and the delivery task exists. Thereafter, the first determiner **120** determines the task ID of a task to be executed (hereinafter, referred to as an execution task) within the acquired task ID. At this time, the first determiner **120** may determine, as the task ID of the execution task, a task ID having the earliest date and time represented by the information correlated with the task ID in the task table of FIG. **10**. Next, the first determiner **120** determines the collection delivery task identified by the determined task ID as an execution task (step S**20** in FIG. **8**).
(97) In the present embodiment, the following description is given by taking, as an example, a case where the execution task is the collection delivery task including the collection task of allowing the delivery vehicle **600** to collect the product I**1** at the shipping location R**1** specified by the orderer and the delivery task of allowing the delivery vehicle **600** to deliver the collected product I**1** to the delivery destination specified by the orderer.
(98) Therefore, the acquirer **110** of the control device **100** acquires the position information of the shipping location R**1** correlated with the task ID of the execution task from the task table of FIG. **10**. Next, the controller **130** generates an acceptance command that includes the acquired position information of the shipping location R**1** and that commands accepting the shipping location R**1** as a movement destination, and outputs the generated acceptance command to the data communication circuit **104** with the delivery vehicle **600** as a destination (step S**21**). In this way, the controller **130** allows the delivery vehicle **600** to accept the shipping location R**1** as a movement destination, and starts executing the collection task of allowing the delivery vehicle **600** to collect the product I**1** at the shipping location R**1** (step S**22**).
(99) Thereafter, the controller **130** the control device **100** determines that the state of the delivery vehicle **600** has been changed from the non-accepting state to the accepting state of accepting the shipping location R**1** as the movement destination (step S**23**), and changes the value of the state flag stored in the information storage **190** to a value indicating the accepting state. Next, the controller **130** outputs a movement start command commanding the start of movement to the accepted movement destination to the data communication circuit **104** with the delivery vehicle **600** as a destination (step S**24**).
(100) Thereafter, the acquirer **110** of the control device **100** attempts to acquire an arrival report from the data communication circuit **104**, and determines whether or not the arrival report has been acquired (step S**25**). When it is determined that the arrival report has not been acquired (step S**25**; No), the acquirer **110** repeats the process of step S**25**.
(101) However, when it is determined that the arrival report has been acquired (step S**25**; Yes), the acquirer **110** of the control device **100** determines that the state of the delivery vehicle **600** has been changed from the state of accepting the shipping location R**1** as the movement destination to the non-accepting state since the delivery vehicle **600** has arrived at the shipping location R**1** (step S**26**). Next, the controller **130** changes the value of the state flag of the information storage **190** to a value indicating the non-accepting state.
(102) Thereafter, the acquirer **110** of the control device **100** acquires, from the task table of FIG. **10**, the product ID and the position information of the shipping location R**1** correlated with the task ID of the execution task, and acquires the store ID of the store ST**1** stored in advance in the information storage **190** in correlation with the position information of the shipping location R**1**. Next, the controller **130** generates a storage request including the acquired product ID and for requesting storage of the prepared product I**1** identified by the product ID in the delivery vehicle **600**. Thereafter, the controller **130** outputs the generated storage request to the data communication circuit **104** with the terminal device **901** of the store S**1** identified by the acquired store ID and

including the shipping location R**1** as a destination (step S**27**).
(103) Thereafter, the acquirer **110** of the control device **100** attempts to acquire a collection completion report from the data communication circuit **104**, and determines whether or not the collection completion report has been acquired (step S**28**). The collection completion report is a report received from the delivery vehicle **600**, and is a report notifying that the collection of the product I**1** at the shipping location R**1** by the delivery vehicle **600** is completed. When it is determined that the collection completion report has not been acquired (step S**28**; No), the acquirer **110** repeats the process of step S**28**.
(104) However, when it is determined that the collection completion report has been acquired (step S**28**; Yes), the acquirer **110** of the control device **100** determines that the execution of the collection task has been ended (step S**29**). Next, in the task table of FIG. **10**, the controller **130** deletes the executed collection task by changing the position information of the shipping location R**1** correlated with the task ID of the execution task to a character string "NULL".
(105) Thereafter, the acquirer **110** of the control device **100** acquires the position information of the delivery destination correlated with the task ID of the execution task from the task table. Thereafter, the controller **130** generates an acceptance command that includes the acquired position information of the delivery destination and that commands accepting the delivery destination as a movement destination. Next, the controller **130** outputs the generated acceptance command to the data communication circuit **104** with the delivery vehicle **600** as a destination (step S**30** in FIG. **9**). In this way, the controller **130** starts executing the delivery task of allowing the delivery vehicle **600** to deliver the collected product I**1** to the delivery destination by allowing the delivery vehicle **600** to accept the delivery destination as a movement destination (step S**31**).
(106) Thereafter, the controller **130** of the control device **100** determines that the state of the delivery vehicle **600** has been changed from the non-accepting state to the accepting state of accepting the delivery destination as the movement destination (step S**32**), and changes the value of the state flag stored in the information storage **190** to a value indicating the accepting state. Next, the controller **130** outputs a movement start command to the data communication circuit **104** with the delivery vehicle **600** as a destination, the movement start command being a command commanding the start of movement to the accepted movement destination (step S**33**).
(107) Thereafter, the acquirer **110** of the control device **100** attempts to acquire an arrival report, and when the acquirer **110** determines that the arrival report has not been acquired (step S**34**; No), the acquirer **110** repeats the process of step S**34**. However, when it is determined that the arrival report has been acquired (step S**34**; Yes), the acquirer **110** determines that the state of the delivery vehicle **600** has been changed from the state of accepting the delivery destination as the movement destination to the non-accepting state since the delivery vehicle **600** has arrived at the delivery destination (step S**35**). Next, the controller **130** changes the value of the state flag of the information storage **190** to a value indicating the non-accepting state.
(108) Thereafter, the acquirer **110** of the control device **100** acquires the user ID and the product ID correlated with the task ID of the execution task from the task table of FIG. **10**, and acquires information representing the name of the product I**1** stored in advance in the information storage **190** in correlation with the acquired product ID. Next, the first determiner **120** determines a password used for authenticating a recipient, who receives the product I**1**, on the basis of a predetermined rule or a software random number. Thereafter, the controller **130** generates a reception request for receiving the product I**1** from the delivery vehicle **600** by inputting the password into the delivery vehicle **600**, the reception request including information representing the determined password and information representing the acquired name of the product I**1**. Next, the controller **130** outputs the generated reception request to the data communication circuit **104** with the terminal device **800** of the orderer identified by the acquired user ID as a destination (step S**36**).
(109) Next, the second determiner **140** of the control device **100** starts counting using a hardware

timer or software timer, which is not illustrated, thereby measuring the elapsed time from the timing when the arrival report is acquired or from the timing when the reception request is output. Thereafter, the acquirer **110** acquires information representing the predetermined time from the information storage **190**. Next, on the basis of whether or not the length of the counted time is equal to or greater than the length of the predetermined time, the controller **130** determines whether or not the predetermined time has elapsed from the start of the counting (step S**37**).
(110) When the length of the counting time is shorter than the length of the predetermined time, the controller **130** of the control device **100** determines that the predetermined time has not elapsed (step S**37**; No). Next, the acquirer **110** attempts to acquire an authentication request from the data communication circuit **104**. The authentication request is a request received from the delivery vehicle **600**, and is a request including information representing the password input into the delivery vehicle **600** by the orderer of the product I**1** and requesting for performing password authentication by using the password.
(111) When the authentication request is not acquired, the acquirer **110** of the control device **100** determines that user authentication is not possible, and repeats the above process from step S**37**. However, when the authentication request is acquired, the acquirer **110** acquires the information representing the password from the authentication request. Next, on the basis of whether or not the password represented by the acquired information matches the password determined by the control device **100**, the controller **130** determines whether or not the password authentication is successful (step S**38**). When the controller **130** determines that the password authentication is not successful and has failed because theses passwords do not match each other (step S**38**; No), the controller **130** outputs a lock maintenance command for maintaining the lock of the door **621***a* included in the delivery vehicle **600** to the data communication circuit **104** with the delivery vehicle **600** as a destination. Thereafter, the control device **100** repeats the above process from step S**37**.
(112) However, when the controller **130** of the control device **100** determines that the password authentication is successful because the two passwords match each other (step S**38**; Yes), the controller **130** outputs an unlocking command that commands unlocking the door **621***a* included in the delivery vehicle **600** to the data communication circuit **104** with the delivery vehicle **600** as the destination. Next, the acquirer **110** of the control device **100** attempts to acquire a delivery completion report from the data communication circuit **104**, and determines whether or not the delivery completion report has been acquired (step S**39**). The delivery completion report is a report received from the delivery vehicle **600**, and is a report notifying that the delivery of the product is completed because handing over of the product I**1** to the orderer is completed. When it is determined that the delivery completion report has not been acquired (step S**39**; No), the acquirer **110** repeats the process of step S**39**.
(113) However, when it is determined that the delivery completion report has been acquired (step S**39**; Yes), the controller **130** determines that the execution of the delivery task of allowing the delivery vehicle **600** to deliver the product I**1** to the delivery destination has been ended since the product I**1** has been handed over to the orderer at the delivery destination (step S**40**).
(114) On the other hand, in step S**37**, when the length of the counting time is equal to or greater than the length of the predetermined time, the controller **130** determines that the predetermined time has elapsed (step S**37**; Yes). Next, the controller **130** regards that the order for the product I**1** has been canceled, and determines that the execution of the delivery task of allowing the delivery vehicle **600** to deliver the product I**1** has been ended (step S**40**).
(115) After the process of step S**40** is performed, the controller **130** of the control device **100** deletes the executed delivery task by deleting a record storing the task ID of the execution task from the task table of FIG. **10** (step S**41**). Thereafter, the control device **100** repeats the above process from step S**12** of FIG. **7**.
(116) When in step S**12**, the control device **100** attempts to acquire the task ID from the task table of FIG. **10** (step S**12**) and determines that the task ID has not been acquired (step S**13**; No), the

control device **100** determines that no collection delivery task exists. Thereafter, the control device **100** repeats the processes of steps S**14** to S**16**. With this, when the delivery vehicle **600** is located at a point different from the point PD of the objective destination, the control device **100** moves the delivery vehicle **600** in the non-accepting state toward the point PD, the non-accepting state being a state of not accepting the shipping locations R**1** or R**2** or the delivery destination as a movement destination. Thereafter, when it is determined that the arrival report notifying the delivery vehicle **600** is arrived at the point PD has been acquired (step S**17**; Yes), the control device **100** performs the processes of steps S**12** and S**13**. However, when it is determined that the arrival report has not been acquired (step S**17**; No), the control device **100** performs processes similar to those of steps S**12** and S**13** (steps S**18** and S**19**).
(117) When it is determined in step S**13** or S**19** that the task ID has been acquired (step S**13**; Yes or step S**19**; Yes), the control device **100** determines that one or a plurality of collection delivery tasks exist. Thereafter, the control device **100** determines one execution task from the one or plurality of existing collection delivery tasks and executes the determined execution task by performing the processes of steps S**20** to S**41**.
(118) In this way, the control device **100** allows the delivery vehicle **600** to accept the shipping location R**1** or R**2** as a movement destination, and then moves the delivery vehicle **600**, of which state has been changed to the accepting state, to the shipping location R**1** or R**2** that is the movement destination. Next, the control device **100** allows the delivery vehicle **600**, of which state has been changed to the non-accepting state by arriving at the shipping location R**1** or R**2**, to collect a product at the shipping location R**1** or R**2**. Thereafter, the control device **100** allows the delivery vehicle **600** to accept the delivery destination as the movement destination, and then moves the delivery vehicle **600**, of which state has been changed to the accepting state, to the delivery destination that is the movement destination. Next, the control device **100** allows the delivery vehicle **600**, of which state has been changed to the non-accepting state by the delivery vehicle **600** arriving at the delivery destination, to hand over the product to the orderer at the delivery destination. Thereafter, the control device **100** repeats the above process from step S**12**.
(119) When the data communication circuit **694** of the delivery vehicle **600** receives the acceptance command that is a command output in step S**21** of FIG. **8** and commands accepting the shipping location R**1** as a movement destination, the CPU **691** of the delivery vehicle **600** performs a movement destination movement process as illustrated in FIG. **15** in order to accept the movement destination according to the acceptance command and move to the movement destination.
(120) When the execution of the movement destination movement process is started, the CPU **691** of the delivery vehicle **600** acquires the acceptance command from the data communication circuit **694**, and acquires the position information of the shipping location R**1** from the acquired acceptance command (step S**71**). Next, the CPU **691** accepts the shipping location R**1** as a movement destination according to the acceptance command. Thereafter, the CPU **691** shifts the state of the delivery vehicle **600** to the accepting state by changing the value of the state flag stored in advance in the flash memory **693***b* to a value indicating the accepting state of accepting the movement destination (step S**72**). The state flag is a flag representing whether or not the state of the delivery vehicle **600** is an accepting state or a non-accepting state of not accepting the movement destination.
(121) Thereafter, the CPU **691** of the delivery vehicle **600** attempts to acquire the movement start command from the data communication circuit **694**, and determines whether or not the movement start command has been acquired (step S**73**). When it is determined that the movement start command has not been acquired (step S**73**; No), the CPU **691** repeats the process of step S**73**. However, when it is determined that the movement start command has been acquired (step S**73**; Yes), the CPU **691** performs the same processes as those of steps S**52**, S**53**, S**55**, and S**56** of FIG. **13** (steps S**74** to S**77**). With this, the delivery vehicle **600** moves along the movement route to the movement destination at the setting speed, and transmits an arrival report to the control device **100**

when arriving at the movement destination.
(122) Thereafter, the CPU **691** of the delivery vehicle **600** shifts the state of the delivery vehicle **600** to the non-accepting state by changing the value of the state flag from a value representing the accepting state to a value representing the non-accepting state (step S**78**). Thereafter, the CPU **691** ends the execution of the movement destination movement process.
(123) After outputting the acceptance command that commands accepting the shipping location R**1** of the product I**1** as the movement destination in step S**21** of FIG. **8**, the CPU **101** of the control device **100** performs a notification control process as illustrated in FIG. **16** in parallel with the movement control process illustrated in FIG. **7** to FIG. **9**, in order to transmit a notification that the start timing of preparation of the product I**1** has arrived (hereinafter, referred to as a start timing notification).
(124) The information storage **190** of the control device **100** stores in advance a preparation time table as illustrated in FIG. **17**, which is used for the notification control process. A plurality of records is stored in advance in the preparation time table. In each record of the preparation time table, a store ID, a product ID identifying a product sold in the store ST**1** or ST**2** identified by the store ID, and information representing the name of the product are stored in correlation with one another. The product sold in the store ST**1** or ST**2** is a product that can be shipped at the shipping location R**1** or R**2** in the store ST**1** or ST**2**.
(125) Furthermore, in each record of the preparation time table, preparation time information representing a preparation time is stored in advance in correlation with the store ID and the product ID, the preparation time is time required from when a shipping request of a product identified by the product ID is accepted in the store ST**1** or ST**2** identified by the store ID until when preparation of the product is completed and then shipment of the product becomes possible at the shipping location R**1** or R**2** in the store ST**1** or ST**2**.
(126) In the present embodiment, the product includes steak, ramen, hamburger, tomato, and bread. Furthermore, in the present embodiment, the preparation for changing a state of the tomato and a state of the bread to a state in which shipment of the tomato and the bread is possible does not include cooking, but the preparation of the steak, the ramen, and the hamburger includes cooking. Therefore, preparation times for the tomato and the bread are preset to be shorter than those for the steak, the ramen, and the hamburger.
(127) Furthermore, in the present embodiment, the property of the product includes the property that the state of the product changes over time. For example, the properties of the steak, the ramen, and the hamburger include the need for cooking to increase the temperature of the product to a temperature higher than a predetermined temperature (hereinafter, need for heating cooking), and the property that the temperature of a product increased higher than the predetermined temperature by cooking becomes lower than the predetermined temperature over time (hereinafter, the property of cooling). On the other hand, for example, the properties of the tomato and the bread do not include the need for heating cooking and the property of cooling.
(128) Moreover, in the present embodiment, the property of the ramen includes the property that noodles become too soft over time. The fact that noodles become too soft means that the state of noodles changes from a state in which noodles contain an amount of water per unit length less than a predetermined amount of water per unit length to a state in which noodles contain an amount of water per unit length greater than or equal to the predetermined amount of water per unit length. On the other hand, for example, the properties of the tomato, the bread, the hamburger, and the steak do not include the property that noodles become too soft.
(129) The product and the property of the product are not limited thereto, and the product may be soup served cold or ice cream. Furthermore, the property of the product may include the need for cooking or storage to decrease the temperature of the product to a temperature lower than a predetermined temperature (hereinafter, need for cooling cooking or cooling storage), and the property that the temperature of a product decreased lower than the predetermined temperature,

increases to a temperature higher than or equal to the predetermined temperature over time (hereinafter, the property of warming).
(130) When the execution of the notification control process is started, the acquirer **110** of the control device **100** acquires the store ID of the store ST**1** stored in advance in the information storage **190** in correlation with the position information of the shipping location R**1** of which accepting as the movement destination is commanded in step S**21** of FIG. **8**. Next, the acquirer **110** acquires the neighboring region information that is the information stored in step S**03** of FIG. **4** and stored in the information storage **190** in correlation with the store ID of the store ST**1** (step S**81**). With this, the neighboring region N**1** that is determined on the basis of the estimated preparation time in the store S**1** including the shipping location R**1** is specified by using the shipping location R**1** accepted as the movement destination as a reference.
(131) Next, the acquirer **110** of the control device **100** acquires the estimated preparation time information correlated with the store ID of the store ST**1** from the estimated preparation time table of FIG. **6** (step S**82**). Thereafter, the acquirer **110** acquires the product ID correlated with the task ID of the execution task determined in step S**20** of FIG. **8** from the task table of FIG. **10**. Next, the acquirer **110** acquires the preparation time information correlated with the store ID of the store ST**1** and the acquired product ID from the preparation time table of FIG. **17** (step S**83**). With this, the preparation time of the product I**1** in the store ST**1** is specified.
(132) Thereafter, the first determiner **120** of the control device **100** corrects the boundary B**1** of the neighboring region N**1** determined on the basis of the estimated preparation time to a boundary B**1**′ as illustrated in FIG. **2** on the basis of the preparation time of the product (hereinafter, also referred to as a shipping product) I**1** requested to be shipped (step S**84**). With this, the first determiner **120** determines the boundary B**1**′ of a neighboring region N**1**′ as illustrated in FIG. **2**, in which the time required for the delivery vehicle **600** to move to the shipping location R**1** is equal to or less than the preparation time of the shipping product I**1**.
(133) Therefore, the first determiner **120** of the control device **100** calculates the ratio of the preparation time represented by the preparation time information acquired in step S**83** to the estimated preparation time represented by the estimated preparation time information acquired in step S**82**. In the present embodiment, since the delivery vehicle **600** moves at the preset setting speed, the ratio of a distance D**1**′ in which the delivery vehicle **600** moves in the preparation time to the distance D**1** in which the delivery vehicle **600** moves in the estimated preparation time is equal to that of the preparation time to the estimated preparation time.
(134) Therefore, the first determiner **120** of the control device **100** determines the boundary B**1**′ by reducing or expanding the boundary B**1** by the calculated ratio centering on the shipping location R**1**. Furthermore, the first determiner **120** determines the determined boundary B**1**′ and a region on the shipping location R**1** side from the boundary B**1**′ as the neighboring region N**1**′ where the time required for the delivery vehicle **600** to move to the shipping location R**1** is equal to or less than the preparation time of the shipping product I**1**.
(135) Thereafter, the acquirer **110** of the control device **100** acquires the position information of the delivery vehicle **600** by performing the same process as that of step S**14** of FIG. **7** (step S**85**). Thereafter, on the basis of the acquired position information of the delivery vehicle **600** and information representing the boundary B**1**′ of the neighboring region N**1**′, the second determiner **140** of the control device **100** determines whether or not the delivery vehicle **600** is located on the boundary B**1**′ of the neighboring region N**1**′ or inside the neighboring region N**1**′ (step S**86**). The inside of the neighboring region N**1**′ means a region on the shipping location R**1** side from the boundary B**1**′ of the neighboring region N**1**′.
(136) When it is determined that the delivery vehicle **600** is not located either on the boundary B**1** of the neighboring region N**1**′ or inside the neighboring region N**1**′ (step S**86**; No), the second determiner **140** of the control device **100** determines that the delivery vehicle **600** is located outside the neighboring region N**1**′. The outside of the neighboring region N**1**′ means a region on an

opposite side of the shipping location R**1** from the boundary B**1**′ of the neighboring region N**1**′.
(137) Next, the control device **100** re-acquires the position information of the delivery vehicle **600** (step S**87**), and then determines whether or not the delivery vehicle **600** is located on or inside the boundary B**1**′ of the neighboring region N**1**′ on the basis of the re-acquired position information (step S**88**) by performing the same processes as those of steps S**85** and S**86**. When it is determined that the delivery vehicle **600** is located outside the neighboring region N**1**′ (step S**88**; No), the control device **100** repeats the above process from step S**87**.
(138) However, when it is determined that the delivery vehicle **600** is located on or inside the boundary B**1**′ of the neighboring region N**1**′ (step S**88**; Yes), the second determiner **140** of the control device **100** determines that the delivery vehicle **600** has arrived at the boundary B**1**′ (step S**89**).
(139) When it is determined in step S**86** that the delivery vehicle **600** is located on or inside the boundary B**1**′ of the neighboring region N**1**′ (step S**86**; Yes) or after the process of step S**89** is performed, the controller **130** of the control device **100** outputs a start timing notification to the data communication circuit **104** with the terminal device **901** of the store ST**1** as a destination (step S**90**). Next, after the data communication circuit **104** of the control device **100** transmits the start timing notification to the terminal device **901**, the controller **130** of the control device **100** ends the execution of the notification control process.
(140) The terminal device **800** carried by the orderer, the terminal device **901** of the store ST**1**, and the terminal device **902** of the store ST**2** are, for example, smartphones or tablet-type or notebook-type personal computers. The terminal device **800** of the orderer transmits an order request for accepting an order to the control device **100** according to an operation of the orderer.
(141) Upon receiving the order request transferred by the control device **100** in step S**62** of FIG. **14**, the terminal device **901** of the store ST**1** displays the received order request. When an employee of the store ST**1** visually recognizes the displayed order request, the employee determines whether or not to accept a shipping request of an order requested to be accepted by the order request. When the employee determines to accept the shipping request, the employee accepts the shipping request, and then performs an operation on the terminal device **901** to transmit an acceptance notification notifying that the shipping request has been accepted. The terminal device **901** transmits the acceptance notification to the control device **100** according to the operation.
(142) Thereafter, upon receiving the start timing notification transmitted by the control device **100** in step S**90** of FIG. **16**, the terminal device **901** of the store ST**1** outputs the start timing notification by display or by voice or sound. The employee of the store ST**1** who has confirmed the start timing notification starts preparing the product I**1**.
(143) However, when the employee determines not to accept the shipping request, the employee does not accept the shipping request, does not perform the operation on the terminal device **901** to transmit the acceptance notification, and does not prepare the product I**1**.
(144) Thereafter, upon receiving the storage request transmitted by the control device **100** in step S**27** of FIG. **8**, the terminal device **901** outputs the received storage request by display or by voice or sound. Upon confirming the output storage request, the employee of the store ST**1** stores the prepared product I**1** in the storage box **621** of the delivery vehicle **600** arrived at the shipping location R**1** in the store ST**1**. Thereafter, the employee performs an operation on the input device **695***c* of the delivery vehicle **600** to lock the door **621***a* of the storage box **621**.
(145) When the input device **695***c* of the delivery vehicle **600** outputs a signal corresponding to the operation, the CPU **691** of the delivery vehicle **600** outputs a control signal for locking the door **621***a* to the driving circuit **699**, and then outputs a collection completion report notifying that the collection of the product I**1** is completed to the data communication circuit **694** with the control device **100** as a destination.
(146) When the data communication circuit **694** of the delivery vehicle **600** receives the acceptance command, which commands accepting the delivery destination as the movement destination and is

output in step S**30** of FIG. **9**, from the control device **100**, the CPU **691** of the delivery vehicle **600** performs the movement destination movement process illustrated in FIG. **15** on the basis of the received acceptance command. With this, the delivery vehicle **600** accepts the delivery destination as the movement destination according to the acceptance command, and shifts the state of the delivery vehicle **600** to the accepting state. Next, upon receiving the movement start command, the delivery vehicle **600** starts moving from the shipping location R**1** to the delivery destination. Thereafter, when the delivery vehicle **600** arrives at the delivery destination, the delivery vehicle **600** transmits an arrival report notifying that the delivery vehicle **600** is arrived at the delivery destination to the control device **100**, and then shifts the state of the delivery vehicle **600** to the non-accepting state.

(147) Upon receiving the reception request, which is a request transmitted from the control device **100** having received the arrival report and is transmitted in step S**36** of FIG. **9**, the terminal device **800** of the orderer outputs the received reception request by display or by voice or sound. In the present embodiment, the delivery destination of the product I**1** is an entrance of a condominium where the orderer resides. Upon confirming the reception request displayed on the terminal device **800**, the orderer moves to the entrance and performs an operation for inputting the password included in the receipt request into the input device **695***c* of the delivery vehicle **600** arriving at the entrance. When the input device **695***c* of the delivery vehicle **600** outputs a signal corresponding to the operation, the CPU **691** of the delivery vehicle **600** performs a process of acquiring the input password on the basis of the signal. Next, the CPU **691** outputs an authentication request including the acquired password and for requesting password authentication to the data communication circuit **694** with the control device **100** as a destination.

(148) When the data communication circuit **694** of the delivery vehicle **600** having transmitted the authentication request to the control device **100** receives a lock maintenance command from the control device **100**, the CPU **691** of the delivery vehicle **600** determines that the password authentication has failed, and does not output a control signal for unlocking the door **621***a* to the driving circuit **699**. Therefore, the door **621***a* is kept locked. Thereafter, the CPU **691** of the delivery vehicle **600** displays a message prompting re-input of a password on the display device **695***b*, and then repeats the above process from a process of acquiring an input password.

(149) When the data communication circuit **694** of the delivery vehicle **600** receives an unlocking command from the control device **100**, the CPU **691** of the delivery vehicle **600** determines that the password authentication is successful, and outputs a control signal for unlocking the door **621***a* to the driving circuit **699**. Thereafter, the CPU **691** acquires a signal output from the imaging device **630**, and determines on the basis of an image represented by the acquired signal whether or not the product I**1** has been received from the unlocked storage box **621** at a predetermined cycle. For this purpose, the CPU **691** acquires information representing a template image of a product stored in advance in the flash memory **693***b*, and performs template matching on the basis of the template image represented by the acquired information and the image represented by the acquired signal and generated by the imaging device **630**. Thereafter, when an image region corresponding to the template image is detected from the image generated by the imaging device **630**, the CPU **691** determines that the product I**1** has been received. Next, since handing over of the product I**1** to the orderer is completed, the CPU **691** outputs a delivery completion report notifying that the delivery of the product is completed to the data communication circuit **694** with the control device **100** as a destination.

(150) With such configurations, the control device **100** controls the delivery vehicle **600**. Furthermore, the control device **100** includes the first determiner **120** that determines the point PD of the objective destination (i) by using each of the two shipping locations R**1** and R**2** as a reference and (ii) on the basis of an estimated preparation time from when a shipping request is accepted until when shipment of an estimated product becomes possible at each of the two shipping locations R**1** and R**2**. Moreover, the control device **100** includes the controller **130** that allows the

delivery vehicle **600** to accept the shipping location R**1** or R**2** of a product or a delivery destination of the product as a movement destination so as to set the state of the delivery vehicle **600** to a state of accepting the movement destination. When the state of the delivery vehicle **600** is the state of accepting the movement destination, the controller **130** moves the delivery vehicle **600** to the movement destination. Furthermore, when the state of the delivery vehicle **600** is the state of not accepting the movement destination, the controller **130** moves the delivery vehicle **600** toward the point PD. Furthermore, the product is a package. Therefore, the control device **100** can suppress the difference between the timing when the delivery vehicle **600** arrives at the shipping location R**1** or R**2** and the timing when preparation of the package is completed and shipment of the package becomes possible at the shipping location R**1** or R**2**.
(151) With such configurations, the control device **100** includes the data communication circuit **104** that transmits, to the terminal device **901** at the shipping location R**1**, the start timing notification notifying that the start timing of preparation has arrived when the delivery vehicle **600** arrives at the boundary B**1**′ of the neighboring region N**1**′ determined (i) by using, as a reference, the shipping location R**1** accepted as the movement destination and (ii) on the basis of the preparation time at the shipping location R**1**. Therefore, the control device **100** can notify the shipping location R**1** of the start timing of preparation in order to suppress the difference between the timing when the delivery vehicle **600** arrives at the shipping location R**1** and the timing when preparation of the product is completed and shipment of the package becomes possible at the shipping location R**1**. Therefore, when the preparation of the product is started at the shipping location R**1** at the timing when the start timing is notified, an increase of a preparation waiting time, which is time from when the delivery vehicle **600** arrives at the shipping location R**1** until when shipment of the product becomes possible, can be suppressed, so that a decrease in utilization efficiency of the shipping location R**1** can be suppressed.
(152) In the present embodiment, the utilization efficiency of the shipping location R**1** is calculated by dividing an area of a region that can be used by a person using the shipping location R**1** by an area of the shipping location R**1**. Furthermore, the area of the region that can be used by a person using the shipping location R**1** is, for example, an area excluding the area of a region where an object including the delivery vehicle **600** is placed from the area of the region of the shipping location R**1**, but is limited to thereto.
(153) Since the shipping location R**1** is the position of the entrance of the store ST**1**, the area of the region of the shipping location R**1** includes the area of a region in the entrance of the store ST**1**. Moreover, the area of the region that can be used by a person using the shipping location R**1** includes, for example, an area excluding the area of the region where an object including the delivery vehicle **600** is placed from the area of the region in the entrance of the store ST**1**. Therefore, the control device **100** can suppress an increase in time that the delivery vehicle **600** obstructs or may obstruct the passage of persons entering and exiting the store ST**1** including customers or employees.
(154) Furthermore, the control device **100** can suppress the difference between the timing when the delivery vehicle **600** arrives at the shipping location R**1** or R**2** and the timing when preparation of a product is completed and shipment of the product becomes possible at the shipping location R**1** or R**2**. Therefore, the control device **100** can suppress an increase in an arrival waiting time from when shipment of the product becomes possible at the shipping location R**1** or R**2** until when the delivery vehicle **600** arrives at the shipping location R**1** or R**2**.
(155) Moreover, since the increase in the arrival waiting time can be suppressed, when an ordered product has the property that the state of the product changes over time, the control device **100** can suppress a state change before the start of delivery, the state change being a change of the state of the product to a state different from the state at the end of preparation. That is, for example, the control device **100** can suppress a state change before the start of delivery, the state change being a change of a product having the property that noodles become too soft over time from a state in

which noodles have not become too soft to a state in which noodles have become too soft. Furthermore, for example, the control device **100** can suppress a temperature change before the start of delivery, the temperature change being a change of the temperature of a product having the need for heating cooking and the property of cooling over time from the temperature higher than the predetermined temperature to a temperature lower than or equal to the predetermined temperature.

(156) Furthermore, with such configurations, an estimated preparation time at the shipping location R**1** is the time until when shipment of an estimated product becomes possible, the estimated product being a product of which shipment is estimated to be requested. Furthermore, when a shipping request is actually accepted, the first determiner **120** of the control device **100** corrects the boundary B**1** of the neighboring region N**1** on the basis of the time from when the shipping request is accepted until when shipment of a shipping product becomes possible, the shipping product being a product of which shipment is requested by the shipping request. The data communication circuit **104** sends the start timing notification when the delivery vehicle **600** arrives at the corrected boundary B**1**′. Therefore, even though the estimated product is different from the shipping product, the control device **100** can notify the shipping location R**1** of the start timing of preparation in order to suppress the difference between the timing when the delivery vehicle **600** arrives at the shipping location R**1** and the timing when the preparation of the shipping product is completed at the shipping location R**1**.

(157) With such configurations, the first determiner **120** of the control device **100** determines the point PD of the objective destination so that the region distances L**1** and L**2** to the boundaries B**1** and B**2** of the neighboring regions N**1** and N**2** satisfy a predetermined condition, the neighboring regions N**1** and N**2** determined (i) by using each of the two shipping locations R**1** and R**2** as a reference and (ii) on the basis of the preparation time at each of the two shipping locations R**1** and R**2**.

(158) Therefore, the control device **100** can determine the point PD of the objective destination at a position where an increase in the total arrival waiting time that occurs when an order is confirmed a plurality of times can be suppressed. In the present embodiment, since the predetermined condition is a condition that the sum of the region distance L**1** and the region distance L**2** is minimum, the control device **100** can determine the point PD of the objective destination at a position where the total arrival waiting time can be minimized.

(159) With such configurations, the first determiner **120** of the control device **100** determines the point PD of the objective destination at a position where a timing difference can be suppressed, the timing difference being a difference between the timing when the delivery vehicle **600** arrives at the shipping location R**1** or R**2** and the timing when preparation of the product becomes possible at the shipping location R**1** or R**2**. Furthermore, the controller **130** of the control device **100** performs a control for moving the delivery vehicle **600** to the determined point PD of the objective destination, and performs a control for moving the delivery vehicle **600** from the determined point PD of the objective destination or from a point on the way to the point PD of the objective destination to the shipping location R**1** or R**2**. Moreover, the controller **130** further performs a control for moving the delivery vehicle **600** storing a product from the shipping location R**1** or R**2** to the delivery destination. Therefore, the control device **100** can suppress a decrease in the delivery efficiency of a package that is a product. In the present embodiment, the delivery efficiency of the package is calculated by dividing the total number of a package delivered by the delivery vehicle **600** by the total delivery time from when the delivery vehicle **600** departs from the point PD of the objective destination to when the delivery vehicle **600** returns to the point PD of the objective destination, but it is not limited thereto.

Modification 1 of Embodiment

(160) In the embodiment, the case where the two shipping locations R**1** and R**2** exist has been described; however, the present disclosure is not limited thereto and K shipping locations (K is an

integer of 3 or more) may exist.
(161) That is, the acquirer **110** of the control device **100** may acquire estimated preparation time information for each of the K shipping locations (K is an integer of 3 or more), and the first determiner **120** may acquire the point PD of the objective destination on the basis of the estimated preparation time information acquired for each of the K shipping locations.
(162) Moreover, the control device **100** may determine, for example, at predetermined intervals such as 30 minutes, whether or not each of the K shipping locations can be specified.
(163) For this purpose, the acquirer **110** of the control device **100** according to the present modification outputs, at the predetermined interval, a business inquiry asking whether or not the store ST**1** is open to the data communication circuit **104** with the terminal device **901** of the store ST**1** as a destination, the store ST**1** including the first shipping location R**1**.
(164) An employee of the store ST**1** according to the present modification performs an operation on the terminal device **901** of the store ST**1** to change a value of a flag representing whether or not the store ST**1** is open to a value "TRUE" representing that the store ST**1** is open, at the timing when the store ST**1** is opened. The terminal device **901** changes the value of the flag to the value "TRUE" according to the operation. Furthermore, the employee of the store ST**1** performs an operation on the terminal device **901** to change the value of the flag to a value "FALSE" representing that the store ST**1** is not open, at the timing when the store ST**1** is closed. The terminal device **901** changes the value of the flag to the value "FALSE" according to the operation. Furthermore, upon receiving a business inquiry from the control device **100**, the terminal device **901** returns an in-business notification notifying that the store ST**1** is open when the value of the flag is "TRUE", and returns no in-business notification when the value of the flag is "FALSE".
(165) When the data communication circuit **104** of the control device **100** according to the present modification receives the in-business notification after the business inquiry is transmitted and before a predetermined time elapses from the transmission, the second determiner **140** determines that the first shipping location R**1** can be specified since the store ST**1** is open. However, when the data communication circuit **104** receives no in-business notification before the predetermined time elapses and after the business inquiry is transmitted, the second determiner **140** determines that the first shipping location R**1** is not specifiable since the store ST**1** is not open. Similarly, the control device **100** determines whether or not the second shipping location R**2** and third to K-th shipping locations can be specified.
(166) Thereafter, the first determiner **120** of the control device **100** according to the present modification excludes a shipping location determined not to be specifiable from the K shipping locations, and determines the point PD of the objective destination on the basis of estimated preparation time information acquired for each of the plurality of non-excluded shipping locations.
(167) With such configurations, even though one or more of the K shipping locations are not specifiable due to temporary closure, other than a regular holiday or outside business hours, the point PD of the objective destination can be determined at a position where a decrease in the delivery efficiency of the delivery vehicle **600** can be suppressed.
Modification 2 of Embodiment
(168) In the embodiment, it has been described that the distance D**1** as the basis for determining the neighboring region N**1**, the distance D**2** as the basis for determining the neighboring region N**2**, the region distance L**1** from the point PD of the objective destination to the boundary B**1** of the neighboring region N**1**, and the region distance L**2** from the point PD of the objective destination to the boundary B**2** of the neighboring region N**2** are Euclidean distances in the three-dimensional space. Furthermore, the neighboring regions N**1** and N**2** have been described to be spherical regions.
(169) However, the present disclosure is not limited thereto, and the distances D**1** and D**2** and the region distances L**1** and L**2** may be Euclidean distances in a horizontal plane, the neighboring regions N**1** and N**2** may be circular regions, and the boundary B**1** of the neighboring region N**1** and

the boundary B**2** of the neighboring region N**2** may be circles. Furthermore, the distances D**1** and D**2** and the region distances L**1** and L**2** may be Manhattan distances in the horizontal plane or Manhattan distance in the three-dimensional space. Moreover, the distances D**1** and D**2** and the region distances L**1** and L**2** may be the square of the Euclidean distance or the Manhattan distance.
(170) Moreover, the distances D**1** and D**2** and the region distances L**1** and L**2** may be distances on a route on which the delivery vehicle **600** can move. In such a case, the first determiner **120** of the control device **100** may determine a plurality of routes to the shipping location R**1** by performing the same process as that of step S**52** of FIG. **13** performed by the delivery vehicle **600**, in order to determine the boundary B**1** of the neighboring region N**1** separated from the shipping location R**1** by the distance D**1**. Next, the first determiner **120** may determine a point where a distance on the route from the shipping location R**1** is equal to the distance D**1** with respect to each of the plurality of determined routes, and determine the boundary B**1** having a polyhedral shape or a polygonal shape so that the boundary B**1** includes the point determined for each of the plurality of routes. The first determiner **120** may determine the boundary B**2** of the neighboring region N**2** in the same manner.
(171) Moreover, the first determiner **120** of the control device **100** may determine a plurality of routes from the point PD of the objective destination to the boundary B**1** by performing the same process as that of step S**52** of FIG. **13**, calculate a distance on each of the plurality of determined routes, and set the calculated minimum distance as the region distance L**1**. The first determiner **120** may determine the region distance L**2** in the same manner.
(172) With such configurations, the first determiner **120** of the control device **100** calculates the distance D**1** on the basis of an estimated preparation time, and determines the boundary B**1** having a polyhedral shape or a polygonal shape on the basis of the calculated distance D**1**. Thereafter, when an order is confirmed, the first determiner **120** corrects the boundary B**1** of the neighboring region N**1** determined on the basis of the estimated preparation time to the boundary B**1**′ on the basis of the preparation time of a shipping product targeted for the confirmed order. Therefore, since the control device **100** determines the boundary B**1**′ by correcting the previously determined boundary B**1**, the amount of calculation required for determining the boundary B**1**′ can be reduced as compared with the case of determining the boundary B**1**′, which has a shape more complicated than a spherical shape or a circular shape, such as a polyhedral shape or a polygonal shape, without correcting the boundary B**1**. Furthermore, since the control device **100** determines the boundary B**1**′ by correcting the previously determined boundary B**1**, the time required for determining the boundary B**1**′ can be shortened as compared with the case of determining the boundary B**1**′ without correcting the boundary B**1**.
Modification 3 of Embodiment
(173) In the embodiment, it has been described the first determiner **120** of the control device **100** calculates the time required for the delivery vehicle **600** to move to the shipping location R**1** on the basis of the setting speed of the delivery vehicle **600**. However, the present disclosure is not limited thereto, and the first determiner **120** calculates the time required for the delivery vehicle **600** to move to the shipping location R**1** on the basis of a movement route of the delivery vehicle **600** and traffic conditions on the movement route.
(174) For this purpose, the first determiner **120** of the control device **100** determines a plurality of routes to the shipping location R**1** by performing the same process as that of step S**52** of FIG. **13** performed by the delivery vehicle **600**. Next, the controller **130** generates a reply request that includes information representing a plurality of edges included in the plurality of determined routes and requests for a replay of information representing traffic conditions on each of the plurality of edges. Thereafter, the controller **130** outputs the generated reply request to the data communication circuit **104** with a not-illustrated traffic server as a destination.
(175) In the present modification, the information representing the edge is information representing the latitude, longitude, and altitude of a start point node of the edge, and the latitude, longitude, and

altitude of an end point node of the edge. Furthermore, the traffic condition of the edge includes a passing time required for passing the edge. The passing time of the edge is an average value of times required for one or more of a plurality of delivery vehicles and a plurality of automobiles to pass through the edge, the one or more of a plurality of delivery vehicles and a plurality of automobiles having passed through the edge per unit time, but may be a maximum value of the times or a minimum value of the times.
(176) When the data communication circuit **104** of the control device **100** receives the information indicating the traffic conditions from the traffic server, the acquirer **110** acquires the information from the data communication circuit **104**. Next, on the basis of the passing time of the edge represented by the acquired information, the first determiner **120** determines a point for each of the plurality of determined routes so that the time that is required for the delivery vehicle **600** to move from the point to the shipping location R**1** equals to an estimated preparation time. Thereafter, the first determiner **120** determines the boundary B**1** of the neighboring region N**1** so that the boundary B**1** includes the point determined for each of the plurality of routes.
(177) Similarly, the control device **100** calculates the times required for the delivery vehicle **600** to move to the shipping location R**2** on the basis of a movement route of the delivery vehicle **600** and traffic conditions on the movement routes, and determines the boundary B**2** of the neighboring region N**2** on the basis of the calculated times.
(178) In the present modification, it has been described that the acquirer **110** of the control device **100** acquires the information representing the traffic conditions of the edges received from the not-illustrated traffic server; however, the present disclosure is not limited thereto. The information storage **190** may store a plurality of information representing the edges and a plurality of information representing the traffic conditions of the edges in advance in correlation with each other, and the acquirer **110** may acquire the information representing traffic conditions correlated with each of the information representing the plurality of edges from the information storage **190**.
(179) The information representing the traffic conditions of the edge stored in the information storage **190** of the control device **100** may be information representing an average value of times required for the delivery vehicle **600** to pass the edge in the past, information representing a minimum value of the times, or information representing a maximum value of the times. Furthermore, the information representing the traffic conditions of the edge stored in the information storage **190** may be information representing an average value of times required for one or more of a plurality of delivery vehicles and a plurality of automobiles to pass the edge, the one or more of a plurality of delivery vehicles and a plurality of automobiles having passed the edge per unit time in the past. The information representing the traffic conditions of the edge may be information representing a minimum value of the times or information representing a maximum value of the times.
(180) In the present modification, it has been described that the traffic conditions of the edge include the passing time of the edge; however, the present disclosure is not limited thereto, and the traffic conditions may include the speed limit of the edge. In such a case, the first determiner **120** of the control device **100** may determine, on the basis of the speed limit of the edge represented by the information acquired by the acquirer **110**, a point for each of the plurality of determined routes so that the time that is required for the delivery vehicle **600** to move from the point to the shipping location R**1** at the speed limit equals to an estimated preparation time. Thereafter, the first determiner **120** may determine the boundary B**1** of the neighboring region N**1** so that the boundary B**1** includes the point determined for each of the plurality of routes.
Modification 4 of Embodiment
(181) In the embodiment, it has been described that the first determiner **120** of the control device **100** determines the point PD of the objective destination so that the region distance L**1**, which is the distance from the point PD of the objective destination to the boundary B**1** of the neighboring region N**1**, and the region distance L**2**, which is the distance from the point PD of the objective

destination to the boundary B**2** of the neighboring region N**2**, satisfy the predetermined condition; however, the present disclosure is not limited thereto.
(182) The first determiner **120** of the control device **100** according to the present modification determines the point PD of the objective destination so that a region time T**1** required for the delivery vehicle **600** to move from the point PD of the objective destination to the boundary B**1** of the neighboring region N**1** and a region time T**2** required for the delivery vehicle **600** to move from the point PD of the objective destination to the boundary B**2** of the neighboring region N**2** satisfy a predetermined condition.
(183) In the present modification, the predetermined condition is a condition that the sum of the region time T**1** and the region time T**2** is minimum. Furthermore, in the present modification, the delivery vehicle **600** moves at a preset setting speed. Therefore, at the point PD of the objective destination where the sum of the region distance L**1** from the point PD of the objective destination to the boundary B**1** and the region distance L**2** from the point PD of the objective destination to the boundary B**2** is minimized, the sum of the region time T**1** and the region time T**2** is minimized. On the contrary, at the point PD of the objective destination where the sum of the region time T**1** and the region time T**2** is minimized, the sum of the region distance L**1** and the region distance L**2** is minimized. Therefore, the first determiner **120** of the control device **100** according to the present modification determines a point, where the sum of the region distance L**1** and the region distance L**2** is minimized, as the point PD of the objective destination by using the known algorithm described in the embodiment.
(184) On the other hand, when the delivery vehicle **600** moves at a speed according to traffic conditions, the first determiner **120** of the control device **100** determines a point where the sum of the region distance L**1** and the region distance L**2** is minimized, and then determines the point and J points (J is a natural number), which are separated from the point by a distance equal to or less than a predetermined distance, as candidates for the point PD of the objective destination. Next, the first determiner **120** determines numbers of J+1 candidates for the point PD of the objective destination from No. 1 to No. J+1 on the basis of a predetermined rule or a software random number.
(185) Thereafter, the first determiner **120** of the control device **100** calculates the total time of the region time T**1** and the region time T**2** for the first candidate by the method described in modification 3 of the embodiment, the region time T**1** being the shortest time required for the delivery vehicle **600** to move from the first candidate to the boundary B**1** of the neighboring region N**1**, the region time T**2** being the shortest time required for the delivery vehicle **600** to move from the first candidate to the boundary B**2** of the neighboring region N**2**. Similarly, the first determiner **120** calculates the total time for the second to J+1-th candidates, and determines a candidate, for which the minimum total time is calculated among the calculated J+1 total times, as the point PD of the objective destination.
Modification 5 of Embodiment
(186) In the embodiment, it has been described that the first determiner **120** of the control device **100** determines the point PD of the objective destination so that the region distance L**1** and the region distance L**2** satisfy the predetermined condition. Furthermore, in the embodiment, it has been described that the predetermined condition is a condition that the sum of the region distance L**1** and the region distance L**2** is minimum; however, the present disclosure is not limited thereto. The predetermined condition may be a condition that the point PD of the objective destination is located at a place where the delivery vehicle **600** can wait (hereinafter, referred to as a standby place) and the sum of the region distance L**1** and the region distance L**2** is minimum.
(187) Therefore, the information storage **190** of the control device **100** stores in advance I pieces (I is a natural number) of position information representing the position of the place, where the delivery vehicle **600** can wait, by using latitude, longitude, and altitude. The first determiner **120** determines numbers of the I pieces of position information to be from No. 1 to No. I, and calculates, for a waiting place at a position represented by the first position information, the total

distance of the region distance L**1** from the waiting place to the boundary B**1** and the region distance L**2** from the waiting place to the boundary B**2**. Similarly, the first determiner **120** calculates the total distance for second to I-th position information, and determines a waiting place, where the minimum total distance is calculated among the calculated I total distances, as the point PD of the objective destination.

(188) In the present modification, it has been described that the predetermined condition is a condition that the point PD of the objective destination is located at a place where the delivery vehicle **600** can wait and the sum of the region distance L**1** and the region distance L**2** is minimum; however, the present disclosure is not limited thereto. The predetermined condition may be a condition that the point PD of the objective destination is located at a place where the delivery vehicle **600** can wait and of which a number that is determined on the basis of the sum of the region distance L**1** and the region distance L**2** is from No. 1 to No. N (N is a natural number equal to or less than I) in ascending order.

(189) Therefore, the first determiner **120** of the control device **100** may determine numbers of the I waiting places to be from No. 1 to No. I in ascending order of the total distance, and determine one waiting place, which is selected, on the basis of a predetermined rule or a software random number, among waiting places of which numbers are determined to be equal to or less than N, as the point PD of the objective destination.

(190) In the present modification, it has been described that the first determiner **120** of the control device **100** determines the point PD of the objective destination so that the point PD of the objective destination is located at a place where the delivery vehicle **600** can wait and the sum of the region distance L**1** and the region distance L**2** is minimum; however, the present disclosure is not limited thereto. The first determiner **120** may determine the point PD of the objective destination so that the point PD of the objective destination is located at a place where the delivery vehicle **600** can wait and the sum of the region time T**1** and the region time T**2** is minimized.

(191) Furthermore, the predetermined condition may be a condition that the point PD of the objective destination is located at a place where the delivery vehicle **600** can wait and of which a number that is determined on the basis of the sum of the region time T**1** and the region time T**2** is from No. 1 to No. N (N is a natural number equal to or less than I) in ascending order.

(192) Furthermore, the predetermined condition may be a condition that the sum of the region distance L**1** and the region distance L**2** is shorter than a predetermined distance, or the predetermined condition may be a condition that the sum of the region time T**1** and the region time T**2** is shorter than a predetermined time.

Modification 6 of Embodiment

(193) In the embodiment, it has been described that the predetermined condition is a condition that the sum of the region distance L**1** and the region distance L**2** is minimized; however, the present disclosure is not limited thereto. The predetermined condition may be any condition as long as it is a condition for determining, as the point PD of the objective destination, a point where the sum of the region distance L**1** and the region distance L**2** decreases, in preference to a point where the sum of the region distance L**1** and the region distance L**2** increases.

(194) Furthermore, the predetermined condition may be any condition as long as it is a condition for determining, as the point PD of the objective destination, a point where the sum of the region time T**1** and the region time T**2** decreases, in preference to a point where the sum of the region time T**1** and the region time T**2** increases.

Modification 7 of Embodiment

(195) When the point PD of the objective destination is located inside the neighboring region N**1**, the first determiner **120** of the control device **100** may correct the region distance L**1** to “0” meters. Furthermore, when the point PD of the objective destination is located inside the neighboring region N**2**, the first determiner **120** may correct the region distance L**2** to “0” meters.

(196) Correcting the region distance L**1** to “0” meters when the point PD of the objective

destination is located inside the neighboring region N**1** is because when an estimated product is the same as a shipping product targeted for a confirmed order and the shipping location R**1** is specified in the order, as long as the point PD of the objective destination is located inside the neighboring region N**1**, the delivery vehicle **600** arrives at the shipping location R**1** before preparation of the product is completed at the shipping location R**1** no matter where the point PD of the objective destination is located. Therefore, in a case where the point PD of the objective destination is located inside the neighboring region N**1**, when the point PD of the objective destination is determined as a point, where an arrival waiting time occurring when specifying the shipping location R**2** is minimized, on the basis of the region distance L**2**, a decrease in the delivery efficiency of the delivery vehicle **600** can be reliably suppressed. Similarly, correcting the region distance L**2** to "0" meters when the point PD of the objective destination is located inside the neighboring region N**2** is because a decrease in the delivery efficiency of the delivery vehicle **600** can be reliably suppressed when the point PD of the objective destination is determined as a point, where an arrival waiting time occurring when specifying the shipping location R**1** is minimized, on the basis of the region distance L**1**.

(197) Furthermore, when the point PD of the objective destination is located inside the neighboring region N**1**, the first determiner **120** of the control device **100** may correct the region distance L**1** to a shorter distance than before the correction, and when the point PD of the objective destination is located inside the neighboring region N**2**, the first determiner **120** of the control device **100** may correct the region distance L**2** to a shorter distance than before the correction.

(198) Moreover, when the point PD of the objective destination is located outside the neighboring region N**1**, the first determiner **120** of the control device **100** may correct the region distance L**1** to "0" meters, and when the point PD of the objective destination is located outside the neighboring region N**2**, the first determiner **120** of the control device **100** may correct the region distance L**2** to "0" meters.

(199) Correcting the region distance L**1** to "0" meters when the point PD of the objective destination is located outside the neighboring region N**1** is because when an estimated product is the same as a shipping product and the shipping location R**1** is specified, as long as the point PD of the objective destination is located outside the neighboring region N**1**, the delivery vehicle **600** arrives at the shipping location R**1** after preparation of the product is completed at the shipping location R**1** no matter where the point PD of the objective destination is located. Therefore, in a case where the point PD of the objective destination is located outside the neighboring region N**1**, when the point PD of the objective destination is determined as a point, where a preparation waiting time occurring when specifying the shipping location R**2** is minimized, on the basis of the region distance L**2**, a decrease in utilization efficiency of the shipping locations R**1** and R**2** can be reliably suppressed.

(200) Furthermore, when the point PD of the objective destination is located outside the neighboring region N**1**, the first determiner **120** of the control device **100** may correct the region distance L**1** to a shorter distance than before the correction, and when the point PD of the objective destination is located outside the neighboring region N**2**, the first determiner **120** of the control device **100** may correct the region distance L**2** to a shorter distance than before the correction.

Modification 8 of Embodiment

(201) In the embodiment, it has been described that the predetermined condition is a condition that the sum of the region distance L**1** and the region distance L**2** is minimized; however, the present disclosure is not limited thereto. The predetermined condition according to the present modification is a condition for minimizing an expected value of a region distance, which is the sum of a weighted region distance L**1** and a weighted region distance L**2**, the weighted region distance L**1** being the region distance L**1** weighted by the probability that an order specifying the first shipping location R**1** is confirmed (hereinafter, referred to as first probability), the weighted region distance L**2** being the region distance L**2** weighted by the probability that an order specifying the second

shipping location R**2** is confirmed (hereinafter, referred to as second probability).
(202) The information storage **190** of the control device **100** according to the present modification stores in advance total confirmation frequency information representing the number of times by which an order has been confirmed. Furthermore, the information storage **190** stores in advance the store ID and confirmation frequency information in correlation with each other, the confirmation frequency information representing the number of times by which an order specifying the shipping location R**1** or R**2** in the store ST**1** or ST**2** identified by the store ID has been confirmed.
(203) When it is determined in step S**63** of FIG. **14** that the order has been accepted (step S**63**; Yes), the second determiner **140** of the control device **100** determines that the order has been confirmed. Thereafter, the acquirer **110** acquires the total confirmation frequency information stored in the information storage **190**, and the controller **130** increments the number of times represented by the acquired total confirmation frequency information by a value “1” and then changes the total confirmation frequency information stored in the information storage **190** into total confirmation frequency information representing the incremented number of times.
(204) Furthermore, the acquirer **110** of the control device **100** acquires the confirmation frequency information stored in the information storage **190** in correlation with the store ID acquired in step S**61**. Next, the controller **130** increments the number of times represented by the acquired confirmation frequency information by a value “1”, and then changes the confirmation frequency information stored in the information storage **190** in correlation with the store ID acquired in step S**61** into confirmation frequency information representing the incremented number of times.
(205) Moreover, in step S**04** of FIG. **4**, the acquirer **110** of the control device **100** acquires the total confirmation frequency information stored in the information storage **190**. Furthermore, the acquirer **110** acquires, from the information storage **190**, confirmation frequency information (hereinafter, referred to as first confirmation frequency information) correlated with the store ID of the store ST**1** including the first shipping location R**1**. Similarly, the acquirer **110** acquires confirmation frequency information (hereinafter, referred to as second confirmation frequency information) correlated with the store ID of the store ST**2** including the second shipping location R**2**.
(206) Thereafter, the acquirer **110** of the control device **100** acquires the first probability by dividing the number of times represented by the first confirmation frequency information by the number of times represented by the total confirmation frequency information. Similarly, the acquirer **110** acquires the second probability by dividing the number of times represented by the second confirmation frequency information by the number of times represented by the total confirmation frequency information.
(207) Next, the first determiner **120** of the control device **100** determines the point PD of the objective destination, where the sum of the region distance L**1** weighted by the first probability and the region distance L**2** weighted by the second probability is minimized, by using, for example, a known algorithm including a least squares method.
(208) It has been described that the predetermined condition according to the present modification is a condition for minimizing the expected value of the region distance, which is the sum of the region distance L**1** weighted by the first probability and the region distance L**2** weighted by the second probability, and the first determiner **120** of the control device **100** determines the point PD of the objective destination where the expected value of the region distance is minimized. However, the present disclosure is not limited thereto. The predetermined condition may be a condition for minimizing an expected value of a region time, which is the sum of the region time T**1** weighted by the first probability and the region time T**2** weighted by the second probability, and the first determiner **120** of the control device **100** may determine the point PD of the objective destination where the expected value of the region time is minimized.
Modification 9 of Embodiment
(209) In the embodiment, it has been described that the estimated product in the store ST**1** is a

product with the highest probability that an order is confirmed among a plurality of products sold in the store ST**1**; however, the present disclosure is not limited thereto.
(210) The estimated product in the store ST**1** may be one product selected from one or a plurality of products with a higher probability that an order is confirmed than a predetermined probability among the plurality of products sold in the store ST**1**, on the basis of a predetermined rule or a software random number. Similarly, the estimated product in the store ST**2** may be one product selected from one or a plurality of products with a higher probability that an order is confirmed than a predetermined probability among the plurality of products sold in the store ST**2**, on the basis of a predetermined rule or a software random number.
(211) Furthermore, the estimated product in the store ST**1** may be one product selected from one or a plurality of products, numbers of which are from No. 1 to a predetermined number in descending order of probability that an order is confirmed, on the basis of a predetermined rule or a software random number, the products being sold in the store ST**1**. Similarly, the estimated product in the store ST**2** may be one product selected from one or a plurality of products, numbers of which are from No. 1 to a predetermined number in descending order of probability that an order is confirmed, on the basis of a predetermined rule or a software random number, the products being sold in the store ST**2**.
(212) Moreover, the estimated product in the store ST**1** may be a product having the longest preparation time or a product having the shortest preparation time among the plurality of products sold in the store ST**1**. Similarly, the estimated product in the store ST**2** may be a product having the longest preparation time or a product having the shortest preparation time among the plurality of products sold in the store ST**2**.
(213) Furthermore, the estimated product in the store ST**1** may be a real product or a virtual product having the length of a preparation time equal to an average value of preparation times of the plurality of products sold in the store ST**1**, or a real product having a preparation time closest to the average value. Similarly, the estimated product in the store ST**2** may be a real product or a virtual product having the length of a preparation time equal to an average value of preparation times of the plurality of products sold in the store ST**2**, or a real product having a preparation time closest to the average value. The average value of the preparation times of the plurality of products may be a value obtained by dividing the sum of the preparation times of the plurality of products by the number of the plurality of products.
(214) Moreover, the estimated product in the store ST**1** may be a real product or a virtual product having the length of a preparation time equal to an expected value of the preparation times of the plurality of products sold in the store ST**1**, or a real product having a preparation time closest to the expected value. That is, when M products (M is a natural number) are sold in the store ST**1**, the length of the preparation time of the estimated product may be the sum of weighted preparation times of first to M-th products. The weighted preparation time of the first product is a preparation time of the first product weighted by the probability that an order for the first product is confirmed, and the weighted preparation times of the second to M-th products are preparation times of the second to M-th products weighted by the probability that an order for the second to M-th products is confirmed. Similarly, the estimated product in the store ST**2** may be a real product or a virtual product having the length of a preparation time equal to an expected value of the preparation times of the plurality of products sold in the store ST**2**, or a real product having a preparation time closest to the expected value.
Modification 10 of Embodiment
(215) In the embodiment, the first determiner **120** of the control device **100** may correct the estimated preparation time of the store ST**1** on the basis of the number of employees attending and working in the store ST**1** (hereinafter, referred to as the number of workers) and determine the neighboring region N**1** on the basis of the corrected estimated preparation time.
(216) For this purpose, among the employees who are attending and working in the store ST**1**, an

employee who is in managerial positions performs an operation on the terminal device **901** of the store ST**1**, for example, at a predetermined time to input the number of workers, and the terminal device **901** generates information representing the number of workers according to the operation and stores the generated information. Thereafter, upon receiving a report request for reporting the number of workers from the control device **100**, the terminal device **901** returns the information indicating the number of workers to the control device **100**.

(217) In step S**01** of FIG. **4**, the acquirer **110** of the control device **100** acquires the estimated preparation time information of the store ST**1** from the estimated preparation time table of FIG. **6**, and then outputs the report request to the data communication circuit **104** with the terminal device **901** of the store ST**1** as a destination. Thereafter, after transmitting the report request to the terminal device **901**, when the data communication circuit **104** of the control device **100** receives the information indicating the number of workers from the terminal device **901**, the acquirer **110** acquires the information from the data communication circuit **104**.

(218) Next, on the basis of the number of workers in the store ST**1** represented by the acquired information, the first determiner **120** of the control device **100** corrects the estimated preparation time of the store ST**1** represented by the estimated preparation time information. For this purpose, the acquirer **110** acquires information representing the reference number of people stored in advance in the information storage **190** in correlation with the store ID of the store ST**1**. Next, for example, by multiplying an absolute value of the difference between the number of workers in the store ST**1** and the reference number of people in the store ST**1** by a positive constant C**1**, the first determiner **120** determines a larger correction value as the absolute value of the difference increases. A person skilled in the art can experimentally determine a suitable value of the constant C**1**.

(219) Thereafter, when the number of workers is greater than the reference number of people, the first determiner **120** of the control device **100** corrects the estimated preparation time to a shorter time than before the correction by subtracting the determined positive correction value from the estimated preparation time. However, when the number of workers is smaller than the reference number of people, the first determiner **120** corrects the estimated preparation time to a longer time than before the correction by adding the determined positive correction value to the estimated preparation time. Furthermore, when the number of workers is equal to the reference number of people, the first determiner **120** corrects the estimated preparation time to the same time as before the correction. Thereafter, the first determiner **120** determines the neighboring region N**1** on the basis of the corrected estimated preparation time of the store ST**1**.

(220) Furthermore, the first determiner **120** of the control device **100** corrects the estimated preparation time of the store ST**2** on the basis of the number of workers in the store ST**2**, and determines the neighboring region N**2** on the basis of the corrected estimated preparation time of the store ST**2**.

(221) Moreover, in step S**82** of FIG. **16**, the first determiner **120** of the control device **100** corrects the estimated preparation time of the store ST**1** on the basis of the number of workers in the store ST**1**, and corrects the estimated preparation time of the store ST**2** on the basis of the number of workers in the store ST**2**. Moreover, in step S**83**, the first determiner **120** corrects the preparation time of the store ST**1** on the basis of the number of workers in the store ST**1**, and corrects the preparation time of the store ST**2** on the basis of the number of workers in the store ST**2**. Thereafter, in step S**84**, the first determiner **120** corrects the boundary B**1** of the neighboring region N**1** on the basis of the corrected estimated preparation time of the store ST**1** and the corrected preparation time of the store ST**1**. Similarly, the first determiner **120** corrects the boundary B**2** of the neighboring region N**2** on the basis of the corrected estimated preparation time of the store ST**2** and the corrected preparation time of the store ST**2**.

Modification 11 of Embodiment

(222) In modification 10 of the embodiment, it has been described that the first determiner **120** of

the control device **100** corrects the preparation time and the estimated preparation time of the store ST**1** on the basis of the number of workers in the store ST**1**; however, the present disclosure is not limited thereto. The first determiner **120** according to the present modification corrects the preparation time and the estimated preparation time of the store ST**1** on the basis of the number of products for which an order has been confirmed but preparation has not been completed in the store ST**1** (hereinafter, referred to as the number of unprepared products). The confirmed order may or may not include an order of a customer who visited the store ST**1**.
(223) For this purpose, when the terminal device **901** of the store ST**1** according to the present modification is started, the terminal device **901** initializes the value of a counter representing the number of unprepared products to a value "0". Next, upon accepting an order, an employee of the store ST**1** performs an operation on the terminal device **901** to increase the value of the counter by a value "1", and the terminal device **901** increases the value of the counter by the value "1" according to the operation. Furthermore, when finishing the preparation of an accepted product, the employee performs an operation of reducing the value of the counter by the value "1", and the terminal device **901** reduces the value of the counter by the value "1" according to the operation. Upon receiving a report request for reporting the number of unprepared products from the control device **100**, the terminal device **901** returns information representing the value of the counter to the control device **100**.
(224) The acquirer **110** of the control device **100** according to the present modification outputs the report request to the data communication circuit **104** with the terminal device **901** of the store ST**1** as a destination. Thereafter, when the data communication circuit **104** of the control device **100** receives information indicating the number of unprepared products, the acquirer **110** acquires the information from the data communication circuit **104**. Next, the first determiner **120** corrects the preparation time and the estimated preparation time of the store ST**1** on the basis of the number of products represented by the acquired information.
(225) For this purpose, the first determiner **120** of the control device **100** determines a larger correction value as the number of products increases by multiplying the number of unprepared products in the store ST**1** by a positive constant C**2**, for example. A person skilled in the art can experimentally determine a suitable value of the constant C**2**.
(226) Thereafter, the first determiner **120** of the control device **100** corrects the preparation time and the estimated preparation time to longer times than before the correction by adding the determined positive correction value to the preparation time and the estimated preparation time, respectively.
(227) Furthermore, the first determiner **120** of the control device **100** corrects the preparation time and the estimated preparation time of the store ST**2** on the basis of the number of unprepared products in the store ST**2**.
Modification 12 of Embodiment
(228) The control device **100** according to modification 10 of the embodiment, which corrects the preparation time and the estimated preparation time of the store ST**1** on the basis of the number of workers in the store ST**1**, and the control device **100** according to modification 11 of the embodiment, which corrects the preparation time and the estimated preparation time of the store ST**1** on the basis of the number of unprepared products in the store ST**1**, can be combined. Therefore, the first determiner **120** of the control device **100** according to the present modification corrects the preparation time and the estimated preparation time of the store ST**1** on the basis of both the number of workers in the store ST**1** and the number of unprepared products in the store ST**1**. Furthermore, the first determiner **120** according to the present modification corrects the preparation time and the estimated preparation time of the store ST**2** on the basis of both the number of workers in the store ST**2** and the number of unprepared products in the store ST**2**.
Modification 13 of Embodiment
(229) On the basis of a response time from when a start timing notification is transmitted to the

terminal device **901** of the store S**1** to when a response transmitted from the terminal device **901** is received, the control device **100** may perform a control for delaying an arrival time of the delivery vehicle **600** to the shipping location R**1**.

(230) For this purpose, upon receiving the start timing notification from the control device **100**, the terminal device **901** of the store ST**1** including the shipping location R**1** outputs the start timing notification by display or by voice or sound. In the present modification, the following description is given by taking, as a specific example, a case where, for example, since an employee of the store ST**1** is preparing a product ordered by a customer who visited the store ST**1**, the start timing notification is confirmed at the timing when "5 minutes" has elapsed from the timing when the start timing notification is output by display or by voice or sound.

(231) Upon confirming the start timing notification, the employee of the store ST**1** performs an operation on the terminal device **901** to transmit a notification notifying that the preparation of the ordered product is started as a response to the start timing notification. When the operation is performed, the terminal device **901** transmits the response to the control device **100**.

(232) When the controller **130** of the control device **100** according to the present modification outputs the start timing notification to the data communication circuit **104** with the terminal device **901** of the store ST**1** as a destination in step S**90** of FIG. **16**, the acquirer **110** starts counting by using a hardware timer or software timer, which is not illustrated.

(233) When the data communication circuit **104** of the control device **100** transmits the start timing notification to the terminal device **901** and then receives a response from the terminal device **901**, the acquirer **110** of the control device **100** acquires time "5 minutes" counted from the transmission of the start timing notification to the reception of the response as a response time.

(234) Thereafter, in the present modification, the first determiner **120** of the control device **100** determines a delay time for delaying an arrival time of the delivery vehicle **600** at the shipping location R**1** as the response time "5 minutes". However, the present disclosure is not limited thereto, and as long as the delay time is based on the response time "5 minutes", the first determiner **120** may determine the length of the delay time to any length. For example, the first determiner **120** may determine the length of the delay time to be longer than the response time "5 minutes" or shorter than the response time "5 minutes".

(235) Next, the controller **130** of the control device **100** generates a route change command that includes information representing the delay time "5 minutes" and that commands changing the movement route of the delivery vehicle **600** to a route (hereinafter, referred to as a delay route) for delaying, by "5 minutes", an arrival time of the delivery vehicle **600** to the shipping location R**1** when the delivery vehicle **600** continues to move along a route on which the delivery vehicle **600** is moving. Thereafter, by outputting the generated route change command to the data communication circuit **104** with the delivery vehicle **600** as a destination, the controller **130** performs a delay control for delaying the arrival time of the delivery vehicle **600** to the shipping location R**1** and then ending the execution of the notification control process.

(236) When the data communication circuit **694** receives the route change command while the delivery vehicle **600** is moving to the shipping location R**1** by performing the movement destination movement process of FIG. **15**, the CPU **691** of the delivery vehicle **600** acquires the route change command from the data communication circuit **694**. Next, the CPU **691** specifies the position of the delivery vehicle **600** on the basis of a signal output from the position measurement circuit **696**, and calculates a movement distance from the position of the delivery vehicle **600** to the shipping location R**1** on the route, on which the delivery vehicle **600** is moving, on the basis of information representing the movement route. Thereafter, the CPU **691** acquires information representing the setting speed stored in the flash memory **693***b*, and divides the movement distance by the setting speed represented by the acquired information. With this, the CPU **691** calculates the movement time required to move from the position of the delivery vehicle **600** to the shipping location R**1** when the delivery vehicle **600** continues to move along the route on which the delivery

vehicle **600** is moving.
(237) Next, the CPU **691** of the delivery vehicle **600** acquires the information representing the delay time "5 minutes" from the route change command and then searches for a plurality of routes from the position of the delivery vehicle **600** to the shipping location R**1**. Thereafter, the CPU **691** determines a route, where the time required for movement is longer by the delay time "5 minutes" than the calculated movement time among the plurality of searched routes, as a delay route. Next, the CPU **691** changes the movement route to the delay route and continues the execution of the movement destination movement process of FIG. **15**. With this, the delivery vehicle **600** arrives at the shipping location R**1** (i) at the timing delayed by the delay time "5 minutes" from the timing when the delivery vehicle **600** arrives at the shipping location R**1** when the delivery vehicle **600** continues to move along the movement route before the change, and (ii) at the timing when preparation of an ordered product is completed at the shipping location R**1**.
(238) With such configurations, the data communication circuit **104** of the control device **100** receives the response to the start timing notification from the terminal device **901** at the shipping location R**1**, and the controller **130** performs a control for delaying the arrival time of the delivery vehicle **600** to the shipping location R**1** on the basis of a response time from the transmission of the start timing notification to the reception of the response. Therefore, the control device **100** can reliably suppress the difference between the timing when the delivery vehicle **600** arrives at the shipping location R**1** and the timing when preparation of a product that is a package is completed and shipment of the product becomes possible at the shipping location R**1**. Therefore, the control device **100** can reliably suppress a decrease in the utilization efficiency of the shipping location R**1**.
(239) In the present modification, it has been described that the control device **100** performs a control for delaying the arrival time of the delivery vehicle **600** to the shipping location R**1** in the store ST**1** on the basis of the response time of the store ST**1**; however, the present disclosure is not limited thereto. The data communication circuit **104** of the control device **100** may transmit the start timing notification to the terminal device **902** of the store ST**2** and then receive a response from the terminal device **902**, and the controller **130** may perform a control for delaying the arrival time of the delivery vehicle **600** to the shipping location R**2** in the store ST**2** on the basis of a response time from the transmission of the start timing notification to the reception of a response.
Modification 14 of Embodiment
(240) In modification 13 of the embodiment, it has been described that the control device **100** controls the movement route of the delivery vehicle **600** on the basis of the response time; however, the present disclosure is not limited thereto. The control device **100** may control the movement speed of the delivery vehicle **600** on the basis of the response time.
(241) For this purpose, the controller **130** of the control device **100** generates a speed change command that includes information representing the delay time "5 minutes" and that commands changing the movement speed of the delivery vehicle **600** to a speed (hereinafter, referred to as a delay speed) for delaying, by the delay time "5 minutes", an arrival time of the delivery vehicle **600** to the shipping location R**1** when the delivery vehicle **600** continues to move at a preset setting speed. Thereafter, the controller **130** outputs the generated speed change command to the data communication circuit **104** with the delivery vehicle **600** as a destination.
(242) When the data communication circuit **694** receives the speed change command while the delivery vehicle **600** is moving to the shipping location R**1** by performing the movement destination movement process of FIG. **15**, the CPU **691** of the delivery vehicle **600** acquires the speed change command from the data communication circuit **694**. Next, the CPU **691** acquires the information representing the delay time "5 minutes" from the acquired speed change command. Furthermore, the CPU **691** calculates a movement distance from the position of the delivery vehicle **600** to the shipping location R**1**. Moreover, the CPU **691** acquires information representing the setting speed of the delivery vehicle **600** from the flash memory **693***b*. Thereafter, on the basis of the calculated movement distance, the delay time "5 minutes", and the setting speed, the CPU **691**

determines a deceleration speed for delaying, by the delay time “5 minutes”, an arrival time of the delivery vehicle **600** to the shipping location R**1**.

(243) Next, the CPU **691** of the delivery vehicle **600** calculates a delay speed slower than the setting speed by the deceleration speed. Thereafter, the CPU **691** outputs a control signal for moving at the calculated delay speed to the driving circuit **699** in the process of step S**75** of FIG. **15**. With this, the delivery vehicle **600** moves to the shipping location R**1** by changing the movement speed from the setting speed to the delay speed.

(244) In the present modification, it has been described that the control device **100** controls the moving speed of the delivery vehicle **600** moving to the shipping location R**1** in the store ST**1** on the basis of the response time of the store ST**1**; however, the present disclosure is not limited thereto. The control device **100** may control the moving speed of the delivery vehicle **600** moving to the shipping location R**2** in the store ST**2** on the basis of the response time of the store ST**2**.

(245) In the present modification, it has been described that the control device **100** controls the moving speed of the delivery vehicle **600** on the basis of the response time; however, the present disclosure is not limited thereto. The control device **100** may perform a control for stopping the delivery vehicle **600** moving to the shipping location R**1** or R**2** by a stop time determined on the basis of the response time. The length of the stop time may be any length as long as it is based on the response time, and may be, for example, the same as that of the response time, may be shorter than that of the response time, or may be longer than that of the response time.

Modification 15 of Embodiment

(246) The control device **100** according to modification 13 of the embodiment and the control device **100** according to modification 14 of the embodiment can be combined, the control device **100** according to modification 13 of the embodiment being a device that performs a control for changing, on the basis of the response time of the store ST**1** or ST**2**, the movement route of the delivery vehicle **600** moving to the shipping location R**1** or R**2** in the store ST**1** or ST**2**, the control device **100** according to modification 14 of the embodiment being a device that performs a control for changing, on the basis of the response time, the moving speed of the delivery vehicle **600** or a control for stopping the delivery vehicle **600** on the basis of the response time. The controller **130** of the control device **100** according to the present modification performs one or more of the control for changing, on the basis of the response time of the store ST**1** or ST**2**, the movement route of the delivery vehicle **600** moving to the shipping location R**1** or R**2** in the store ST**1** or ST**2**, the control for changing the moving speed on the basis of the response time, and the control for stopping the delivery vehicle **600** on the basis of the response time.

Modification 16 of Embodiment

(247) Although FIG. **2** referenced in the description of the embodiment illustrates the point PD of the objective destination located outside the neighboring regions N**1** and N**2**, the position of the point PD of the objective destination determined by the first determiner **120** of the control device **100** is not limited to the outside of the neighboring regions N**1** and N**2**. For example, as illustrated in FIG. **18**, when the neighboring region N**2** is included in the neighboring region N**1** and there is no point of tangency between the boundary B**1** of the neighboring region N**1** and the boundary B**2** of the neighboring region N**2**, the first determiner **120** determines the point PD of the objective destination at a position located inside the neighboring region N**1** and located outside the neighboring region N**2**.

(248) Furthermore, as illustrated in FIG. **19**, when there is a point of tangency between the boundary B**1** of the neighboring region N**1** and the boundary B**2** of the neighboring region N**2**, the first determiner **120** of the control device **100** determines the point PD of the objective destination at a position of the point of tangency. Moreover, as illustrated in FIG. **20**, when there are two points of intersection of the boundary B**1** of the neighboring region N**1** and the boundary B**2** of the neighboring region N**2**, the first determiner **120** determines the point PD of the objective destination at a position of one point of intersection selected from the two points of intersection on

the basis of a predetermined rule or a software random number.

Modification 17 of Embodiment

(249) In the embodiment, the case where an order specifying the store ST**1** and the product I**1** sold in the store ST**1** is accepted has been described as a specific example; however, the present disclosure is not limited thereto. An order specifying the store ST**2** and a not-illustrated product **12** sold in the store ST**2** may be accepted.

(250) Furthermore, in the embodiment, the case where the shipping location R**1** in the store ST**1** is specified and the boundary B**1** of the neighboring region N**1** of the shipping location R**1** is corrected to the boundary B**1**′ has been described as a specific example; however, the present disclosure is not limited thereto. The shipping location R**2** in the store ST**2** may be specified and the boundary B**2** of the neighboring region N**2** of the shipping location R**2** may be corrected.

Modification 18 of Embodiment

(251) In the embodiment, it has been described that the first determiner **120** of the control device **100** corrects, on the basis of the preparation time of the shipping product I**1** requested to be shipped, the boundary B**1** of the neighboring region N**1** determined on the basis of the estimated preparation time to the boundary B**1**′ and the second determiner **140** determines whether or not the delivery vehicle **600** has arrived at the corrected boundary B**1**′. However, the present disclosure is not limited thereto. The first determiner **120** may determine a new neighboring region on the basis of the preparation time of the shipping product I**1** without correcting the boundary B**1** of the neighboring region N**1** determined on the basis of the estimated preparation time. Furthermore, the second determiner **140** may determine whether or not the delivery vehicle **600** has arrived at a boundary of the determined new neighboring region.

Modification 19 of Embodiment

(252) In the embodiment, it has been described that the delivery vehicle **600** determines the movement route; however, the present disclosure is not limited thereto, and the first determiner **120** of the control device **100** may determine the movement route of the delivery vehicle **600**. In such a case, the controller **130** of the control device **100** may output a movement command, an acceptance command, or a movement start command including information indicating the determined movement route to the data communication circuit **104** with the delivery vehicle **600** as a destination. Furthermore, the delivery vehicle **600** may move along the movement route represented by the information included in the received movement command, acceptance command, or movement start command.

Modification 20 of Embodiment

(253) In the embodiment, it has been described that when the control device **100** is started, the control device **100** performs the objective destination determination process illustrated in FIG. **4**. However, the present disclosure is not limited thereto. The control device **100** may perform the objective destination determination process at the timing when it is determined in step S**13** of FIG. **7** that the task ID is not acquired (that is, it is determined that there is no collection delivery task), at a predetermined cycle, or at the timing when the input device **105***c* outputs a signal corresponding to an operation performed by an employee of an intermediary.

Modification 21 of Embodiment

(254) In the embodiment, it has been described that the control device **100** performs the objective destination determination process illustrated in FIG. **4** in order to move the delivery vehicle **600** to the point PD of the objective destination and performs the notification control process illustrated in FIG. **16** in order to provide a notification of the start timing of preparation. However, the present disclosure is not limited thereto, and the control device **100** performs the objective destination determination process, but may not perform the notification control process. Furthermore, the control device **100** performs the notification control process, but may not perform the objective destination determination process.

Modification 22 of Embodiment

(255) In the embodiment, it has been described that the delivery system **1** includes the delivery vehicle **600** as an unmanned ground vehicle. However, the present disclosure is not limited thereto, and the delivery system **1** according to the present modification includes, for example, a delivery vehicle **700** as an unmanned aerial vehicle such as a drone as illustrated in FIG. **21**.
(256) The delivery vehicle **700** includes a control device **790** that controls the attitude and flight of the delivery vehicle **700**, propeller arms **701** and **702** protruding from the front surface of the control device **790** to the front right and the front left of the control device **790**, respectively, and propeller arms **703** and **704** protruding from the rear surface of the control device **790** to the rear left and the rear right of the control device **790**, respectively. Furthermore, the delivery vehicle **700** includes propellers **711** to **714** installed at distal ends of the propeller arms **701** to **704**, respectively, and not-illustrated motors for rotating the propellers **711** to **714** under the control of the control device **790**.
(257) Moreover, the delivery vehicle **700** is provided on the lower surface of the control device **790** thereof with a first holding frame **721***a* and a second holding frame **721***b* that surround and hold a product. The first holding frame **721***a* surrounds and holds four sides of one (hereinafter, referred to as a first surrounded surface) of the side surfaces of a rectangular parallelepiped-shaped cardboard in which the product is packaged, and the second holding frame **721***b* surrounds and holds four sides of another side surface (hereinafter, referred to as a second surrounded surface) facing the first surrounded surface surrounded and held by the first holding frame **721***a*.
(258) Furthermore, the delivery vehicle **700** is provided on the lower surface of the control device **790** thereof with guide rails **722***a* and **722***b* that extend in the normal direction of the first surrounded surface and the second surrounded surface of the product, suspend the first holding frame **721***a* and the second holding frame **721***b*, and have the movement direction of the first holding frame **721***a* and the second holding frame **721***b* as an extension direction.
(259) Moreover, the delivery vehicle **700** includes not-illustrated motors that allow the first holding frame **721***a* and the second holding frame **721***b* to surround and hold the product by moving the first holding frame **721***a* and the second holding frame **721***b* in a direction approaching each other under the control of the control device **790**. The not-illustrated motor allows the first holding frame **721***a* and the second holding frame **721***b* to release the surrounded and held product by moving the first holding frame **721***a* and the second holding frame **721***b* in a direction away from each other under the control of the control device **790**. In the present modification, collecting a product by the delivery vehicle **700** means that the delivery vehicle **700** surrounds and holds the product with the first holding frame **721***a* and the second holding frame **721***b*.
(260) The delivery vehicle **700** is provided on the upper surface of the control device **790** thereof with an imaging device **730** having an optical axis and a view angle adjusted so that a recipient is included in an imaging range, the recipient being a person who receives the product released from the first holding frame **721***a* and the second holding frame **721***b*. The configuration and function of the imaging device **730** included in the delivery vehicle **700** are the same as those of the imaging device **630** included in the delivery vehicle **600**.
(261) The delivery vehicle **700** includes support legs **740** that protrude downward from the lower surface of the control device **790** and support the control device **790**.
(262) Furthermore, the delivery vehicle **700** includes a LiDAR sensor **741** provided on the front surface of the control device **790** and a not-illustrated LiDAR sensor provided on the rear surface of the control device **790**. The configurations and functions of the LiDAR sensor **741** on the front surface and the LiDAR sensor on the rear surface included in the delivery vehicle **700** are the same as those of the LiDAR sensor **641** on the front surface and the LiDAR sensor on the rear surface included in the delivery vehicle **600**.
(263) The control device **790** of the delivery vehicle **700** includes a CPU, a RAM, a ROM, a flash memory, a data communication circuit, a video card, a display device, an input device, a position measurement circuit, an input/output port, and a not-illustrated driving circuit, which are hardware.

The configurations and functions of the hardware included in the control device **790** of the delivery vehicle **700** are the same as those of the hardware included in the control device **690** of the delivery vehicle **600** illustrated in FIG. **12**.
(264) The driving circuit of the delivery vehicle **700** is connected to not-illustrated cables connected to the not-illustrated motors for rotating the propellers **711** to **714**. The driving circuit drives the not-illustrated motors for rotating the propellers **711** to **714** according to a signal output by the CPU. Furthermore, the driving circuit of the delivery vehicle **700** drives the not-illustrated motors that move the first holding frame **721***a* and the second holding frame **721***b* according to a signal output by the CPU.
(265) When the not-illustrated data communication circuit included in the delivery vehicle **700** receives a movement command, the CPU of the delivery vehicle **700** performs similar process to the objective destination movement process illustrated in FIG. **13**. With this, the delivery vehicle **700** flies at a setting speed according to the movement command and moves to the destination point PD of the objective destination.
(266) Furthermore, when the data communication circuit included in the delivery vehicle **700** receives an acceptance command, the CPU of the delivery vehicle **700** performs the similar process to the movement destination movement process illustrated in FIG. **15**. With this, the delivery vehicle **700** accepts the shipping location R**1** or R**2** or a delivery destination as a movement destination according to the movement command, and shifts the state of the delivery vehicle to an accepting state. Next, the delivery vehicle **700** flies at the setting speed, moves to the shipping location R**1** or R**2** or the movement destination as the delivery destination, and then shifts the state of the delivery vehicle to a non-accepting state.
(267) In the present modification, it has been described that the delivery vehicle **700** is provided on the lower surface of the control device **790** thereof with the first holding frame **721***a* and the second holding frame **721***b* that surround and hold a product and the guide rails **722***a* and **722***b* that have the movement direction of the first holding frame **721***a* and the second holding frame **721***b*. However, the present disclosure is not limited thereto, and the delivery vehicle **700** may be provided on the lower surface of the control device **790** thereof with a not-illustrated storage cabinet for storing a product. The configuration and function of the storage cabinet included in the delivery vehicle **700** is the same as those of the storage cabinet **620** included in the delivery vehicle **600**.
(268) Furthermore, in the present modification, the delivery vehicle **700** has been described as an unmanned aerial vehicle; however, the present disclosure is not limited thereto, and the delivery vehicle **700** may be an unmanned flying object. Moreover, in the present modification, the delivery vehicle **700** has been described as a drone that obtains lift and thrust with the propellers **711** to **714**; however, the present disclosure is not limited thereto. The delivery vehicle **700** may include wings and obtain lift with the wings, or include an air sac filled with gas having a specific gravity smaller than that of air and obtain lift with the air sac. Furthermore, the delivery vehicle **700** may include a jet engine or a rocket engine, and obtain thrust with the jet engine or the rocket engine.
Modification 23 of Embodiment
(269) In the embodiment, it has been described that the target of an order is a product; however, the present disclosure is not limited thereto. The target of an order may be any object as long as it is a package, and may be, for example, an object that is not the target of commercial transactions. Furthermore, the target of an order may be a living thing instead of an object.
(270) In the embodiment, it has been described that the product is a food or drink; however, the present disclosure is not limited thereto, and the product may be, for example, an object different from food and drink, such as a book or a home appliance.
Modification 24 of Embodiment
(271) In the embodiment, it has been described that the shipping locations R**1** and R**2** are the positions of entrances of the stores ST**1** and ST**2** that sell ordered foods and drinks; however, the

present disclosure is not limited thereto. The shipping locations R**1** and R**2** may be any positions as long as the delivery vehicle **600** can be stopped or parked, or the delivery vehicle **700** can land, and may be, for example, positions inside the stores ST**1** and ST**2**, positions of parking lots of the stores ST**1** and ST**2**, positions in warehouses where ordered products are stored, or positions of shipping doors of the warehouses.

(272) Furthermore, in the embodiment, it has been described that the delivery destination is the entrance of the condominium in which the recipient resides; however, the present disclosure is not limited thereto. The delivery destination may be any position as long as the delivery vehicle **600** can be stopped or parked, or the delivery vehicle **700** can land. Furthermore, the point PD of the objective destination may be any position as long as the delivery vehicle **600** can be stopped or parked, or the delivery vehicle **700** can land.

(273) The position where the delivery vehicle **600** can be stopped or parked may be, for example, the entrance of an apartment house, an office building, a hotel, a commercial facility, or a public facility, or the front door of a house. Furthermore, the position where the delivery vehicle **600** can be stopped or parked may be a lobby of an apartment house, an office building, a hotel, a commercial facility, or a public facility, a garden or a yard of a house, an apartment house, an office building, a hotel, a commercial facility, or a public facility, a parking lot, a river beach, or a park.

(274) In addition to the position where the delivery vehicle **600** can be stopped or parked, the position where the delivery vehicle **700** can land may be a veranda or a rooftop of a house, an apartment building, an office building, a hotel, a commercial facility, or a public facility.

(275) Furthermore, when arriving at the point PD of the objective destination, the delivery vehicle **600** does not stop or park at the point PD of the objective destination, but may circularly move or reciprocate in the vicinity of the point PD of the objective destination at a predetermined speed. The vicinity of the point PD of the objective destination means a region on the side of the point PD of the objective destination with respect to a boundary line separated from the point PD of the objective destination by a predetermined distance. Furthermore, the delivery vehicle **600** may stop once in the vicinity of the point PD of the objective destination and then circularly move or reciprocate in the vicinity of the point PD of the objective destination, may stop after circularly moving or reciprocating, or may repeat the stop and circularly moving or reciprocating.

(276) Similarly, when arriving at the delivery destination or the shipping location R**1** or R**2**, the delivery vehicle **600** may circularly move or reciprocate in the vicinity of the delivery destination or in the vicinity of the shipping location R**1** or R**2** at a predetermined speed. Furthermore, the delivery vehicle **600** may stop once in the vicinity of the delivery destination or in the vicinity of the shipping location R**1** or R**2**, and then may circularly move or reciprocate in the vicinity of the delivery destination or in the vicinity of the shipping location R**1** or R**2**, may stop after circularly moving or reciprocating, or may repeat the stop and circularly moving or reciprocating.

(277) Moreover, when arriving at the point PD of the objective destination, the delivery vehicle **700** does not land at the point PD of the objective destination, and may circularly move or reciprocate in the vicinity of the point PD of the objective destination at a speed equal to or lower than a predetermined speed, or may perform hovering flight at an altitude within a predetermined range. The vicinity of the point PD of the objective destination means an airspace on the point PD of the objective destination side with respect to a boundary surface separated from the point PD of the objective destination by a predetermined distance.

(278) Similarly, when arriving at the delivery destination or the shipping location R**1** or R**2**, the delivery vehicle **700** may circularly move or reciprocate in the vicinity of the delivery destination or in the vicinity of the shipping location R**1** or R**2** at a speed equal to or lower than a predetermined speed, or may perform hovering flight at an altitude within a predetermined range.

Modification 25 of Embodiment

(279) In the embodiment, the delivery vehicle **600** has been described as an unmanned ground vehicle. Furthermore, in modification 22 of the embodiment, the delivery vehicle **700** has been

described as an unmanned aerial vehicle. However, the delivery vehicles **600** and **700** do not necessarily have to be unmanned, and a person may be on board as long as the delivery vehicles **600** and **700** are objects that move autonomously, except for the control by the control device **100**.

Modification 26 of Embodiment

(280) In the embodiment, it has been described that when the delivery vehicle **600** is located at a point different from the point PD of the objective destination in two periods, the control device **100** performs a control (hereinafter, referred to as objective destination movement control) for moving the delivery vehicle **600** toward the point PD.

(281) One of the two periods is a first period from the timing (hereinafter, referred to as starting timing) when the control device **100** is started to the timing when an order is accepted and a collection delivery task is generated for the first time after the startup. Since the control device **100** allows the delivery vehicle **600** to accept the shipping location R**1** or R**2** as a movement destination when the order is accepted and the collection delivery task is generated, the first period may be a period from the start timing to the timing when the control device **100** allows, for the first time after the startup, the delivery vehicle **600** to accept the movement destination.

(282) The remaining one of the two periods is a second period from the timing when all generated collection delivery tasks are deleted to the timing when an order is accepted and a collection delivery task is first generated after all the collection delivery tasks are deleted. Since the control device **100** deletes the collection delivery task when the delivery of the product is completed, the second period may be a period from the timing when the delivery by the delivery vehicle **600** is completed for all delivery destinations specified in the accepted order to the timing when the control device **100** allows, for the first time after the delivery is completed for all the delivery destinations, the delivery vehicle **600** to accept the movement destination.

(283) However, the period during which the control device **100** performs the objective destination movement control is not limited to both the first period and the second period, and may be either the first period or the second period.

(284) Furthermore, in the embodiment, it has been described that the execution of the objective destination movement control is prevented in two periods, that is, the objective destination movement control is not performed in two periods.

(285) One of the two periods is a third period from the timing when a collection delivery task is generated for the first time after the start of the control device **100** to the timing when all generated collection delivery tasks are deleted. Furthermore, the remaining one of the two periods is a fourth period from the timing when a collection delivery task is generated for the first time after all the generated collection delivery tasks are deleted to the timing when all the generated collection delivery tasks are deleted.

(286) When the collection delivery task is generated, the control device **100** allows the shipping location R**1** or R**2** to be accepted as a movement destination, and when the delivery is completed, the control device **100** deletes the collection delivery task. Therefore, the third period may be a period from the timing when the control device **100** allows, for the first time after the startup, the delivery vehicle **600** to accept the movement destination to the timing when the delivery by the delivery vehicle **600** is completed for all delivery destinations. Furthermore, the fourth period may be a period from the timing when the control device **100** allows, for the first time after the delivery for all the delivery destinations is completed, the delivery vehicle **600** to accept the movement destination to the timing when the delivery by the delivery vehicle **600** is completed for all the delivery destinations.

(287) However, the period during which the control device **100** does not perform the objective destination movement control is not limited to both the third period and the fourth period, and may be either the third period or the fourth period.

Modification 27 of Embodiment

(288) In the embodiment, it has been described that the control device **100** stores the state flag and

changes the value of the state flag to a value indicating the non-accepting state when it is determined in step S**11** of FIG. **7**, step S**26** of FIG. **8**, and step S**35** of FIG. **9** that the state of the delivery vehicle **600** is the non-accepting state. However, the present disclosure is not limited thereto, and the control device **100** may not store the state flag and may not perform the processes of steps S**11**, S**26**, and S**35**. Furthermore, in the embodiment, it has been described that when it is determined in steps S**23** and S**32** that the state of the delivery vehicle **600** is the accepting state, the control device **100** changes the value of the state flag to a value indicating the accepting state; however, the present disclosure is not limited thereto. The control device **100** may not store the state flag and may not perform the processes of steps S**23** and S**32**.
(289) Moreover, in the embodiment, it has been described that the delivery vehicle **600** stores the state flag and shifts the state of the delivery vehicle **600** to the accepting state by changing the value of the state flag to a value indicating the accepting state in step S**72** of FIG. **15**. Furthermore, in the embodiment, it has been described that the delivery vehicle **600** shifts the state of the delivery vehicle **600** to the non-accepting state by changing the value of the state flag to a value indicating the non-accepting state in step S**78**. However, the present disclosure is not limited thereto, and the delivery vehicle **600** may not store the state flag and may not perform the processes of steps S**72** and S**78**.
(290) Moreover, in the embodiment, it has been described that the control device **100** outputs the movement start command in step S**24** after outputting the acceptance command in step S**21** of FIG. **8**. Furthermore, it has been described that the control device **100** outputs the movement start command in step S**33** after outputting the acceptance command in step S**30** of FIG. **9**. However, the present disclosure is not limited thereto, and the control device **100** may output the movement start command and the acceptance command at the same time.
(291) Furthermore, the present disclosure is not limited thereto, and the control device **100** may output the acceptance command after outputting the movement start command. In such a case, upon receiving the movement start command, the delivery vehicle **600** may perform the movement destination movement process illustrated in FIG. **15**. At this time, the delivery vehicle **600** may perform a process of determining whether or not the acceptance command has been acquired, instead of the process of step S**73**. When it is determined that the acceptance command has been acquired, the delivery vehicle **600** may acquire position information of a movement destination from the acceptance command by performing the process of step S**71**. Thereafter, the delivery vehicle **600** may continuously perform the movement destination movement process from step S**74**.
(292) Furthermore, the present disclosure is not limited thereto, and instead of the acceptance command and the movement start command, the control device **100** may output a movement command including the position information of a movement destination that is the shipping location R**1** or R**2** or a delivery destination and commanding movement to the movement destination.
Modification 28 of Embodiment
(293) In the embodiment, it has been described that the control device **100** includes the information storage **190**; however, the present disclosure is not limited thereto. The control device **100** according to the present modification does not include the information storage **190**. The control device **100** according to the present modification is connected to a not-illustrated information storage device that is, for example, a network attached storage (NAS) and that has the same function as that of the information storage **190** via the Internet IN, and performs the objective destination determination process illustrated in FIG. **4**, the movement control process illustrated in FIG. **7** to FIG. **9**, the task generation process illustrated in FIG. **14**, and the notification control process illustrated in FIG. **16** by using information stored in the information storage device. The delivery system **1** according to the present modification may or may not include the information storage device.
Modification 29 of Embodiment

(294) In the embodiment, it has been described that the delivery system **1** includes the control device **100**; however, the present disclosure is not limited thereto, and the delivery system **1** may not include the control device **100**. In such a case, the objective destination determination process illustrated in FIG. **4**, the movement control process illustrated in FIG. **7** to FIG. **9**, the task generation process illustrated in FIG. **14**, and the notification control process illustrated in FIG. **16** may be performed by the CPU **691** of the control device **690** included in the delivery vehicle **600**. Therefore, the CPU **691** of the delivery vehicle **600** may serve as not-illustrated functional units corresponding to the acquirer **110**, the first determiner **120**, the controller **130**, and the second determiner **140** of the control device **100**. Furthermore, the flash memory **693***b* of the delivery vehicle **600** may serve as a not-illustrated functional unit corresponding to the information storage **190** of the control device **100**.
(295) The example of the present disclosure and modifications 1 to 29 of the embodiment can be combined with each other. It is of course possible to provide a control device **100** having a configuration for implementing the function according to any one of the embodiment and modifications 1 to 28 of the embodiment and a control device **690** having a configuration for implementing the function according to modification 29 of the embodiment, and it is also possible to provide a system including a plurality of devices and having, as a whole system, a configuration for implementing the function according to any one of the embodiment of the present disclosure and modifications 1 to 29 of the embodiment.
(296) It is possible to provide a control device **100** previously having the configuration for implementing the function according to any one of the embodiment and modifications 1 to 28 of the embodiment. Furthermore, application of a program enables an existing control device to serve as the control device **100** according to any one of the embodiment and modifications 1 to 28 of the embodiment. That is, a computer (CPU or the like) that controls the existing control device executes a program for implementing each functional configuration by the control device **100** exemplified in any one of the embodiment and modifications 1 to 28 of the embodiment, thereby allowing the existing control device to serve as the control device **100** according to any one of the embodiment and modifications 1 to 28 of the embodiment.
(297) It is possible to provide a control device **690** previously having the configuration for implementing the function according to modification 29 of the embodiment of the present disclosure. Furthermore, application of a program enables an existing control device to serve as the control device **690** according to modification 29 of the embodiment. That is, a computer (CPU or the like) that controls the existing control device executes a program for implementing each functional configuration by the control device **690** exemplified in modification 29 of the above example, thereby allowing the existing control device to serve as the control device **690** according to modification 29 of the embodiment.
(298) A distribution method of such a program is arbitrary, and can be stored and distributed in a recording medium such as a memory card, a compact disc (CD)-ROM, or a digital versatile disc (DVD)-ROM, or can also be distributed via a communication medium such as the Internet.
(299) The method according to the present disclosure can be carried out by using the control device **100** according to any one of the embodiment and modifications 1 to 28 of the embodiment and the control device **690** according to modification 29 of the embodiment. Furthermore, the method according to the present disclosure can be carried out by using the delivery system **1** according to any one of the embodiment and modifications 1 to 29 of the embodiment.
(300) The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full

range of equivalents to which such claims are entitled.

Appendices

Appendix 1

(301) A control device for controlling a delivery vehicle, the control device including: a determiner that determines a point (i) by using each of a plurality of shipping locations as a reference and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at each of the plurality of shipping locations; and a controller that allows the delivery vehicle to accept a shipping location of a package or a delivery destination of the package as a movement destination so as to set a state of the delivery vehicle to a state of accepting the movement destination, moves the delivery vehicle to the movement destination when the state of the delivery vehicle is the state of accepting the movement destination, and moves the delivery vehicle toward the point when the state of the delivery vehicle is a state of not accepting the movement destination.

Appendix 2

(302) The control device according to appendix 1, further including: a communicator that transmits, to a terminal device at the shipping location, a notification notifying that a start timing of preparation has arrived when the delivery vehicle arrives at a boundary of a region determined (i) by using, as a reference, the shipping location accepted as the movement destination and (ii) on the basis of the preparation time at the shipping location.

Appendix 3

(303) The control device according to appendix 2, wherein on the basis of a response time from transmission of the notification to reception of a response to the notification, the controller performs a control for delaying an arrival time of the delivery vehicle to the shipping location.

Appendix 4

(304) The control device according to appendix 2 or 3, wherein the preparation time at the shipping location is a time until when shipment of an estimated product becomes possible, the estimated product being a product of which shipment is estimated to be requested, when the shipping request is accepted, the determiner corrects the boundary of the region on the basis of a time from when the shipping request is accepted until when shipment of a shipping product becomes possible, the shipping product being a product of which shipment is requested by the shipping request, and the communicator transmits the notification when the delivery vehicle arrives at the corrected boundary.

Appendix 5

(305) The control device according to any one of appendices **2** to **4**, wherein the determiner determines the point so that distances to the boundary of the region or movement times of the delivery vehicle to the boundary of the region satisfy a predetermined condition, the region determined (i) by using each of the plurality of shipping locations as a reference and (ii) on the basis of the preparation time at each of the plurality of shipping locations.

Appendix 6

(306) The control device according to appendix 5, wherein the determiner acquires a probability that the shipping request is accepted for each of the plurality of shipping locations, and determines the point so that the distances or the movement times weighted by the acquired probability satisfy the predetermined condition.

Appendix 7

(307) The control device according to appendix 6, wherein the predetermined condition is a condition that a sum of the weighted distances or the weighted movement times is minimized.

Appendix 8

(308) The control device according to any one of appendices **2** to **7**, wherein the point includes a point inside the region.

Appendix 9

(309) The control device according to any one of appendices **2** to **8**, wherein the determiner corrects the preparation time at each of the plurality of shipping locations on the basis of at least one of the number of employees on duty at each of the plurality of shipping locations or the number of packages for each of which the shipping request is accepted but preparation is not completed.

Appendix 10

(310) The control device according to any one of appendices **1** to **9**, wherein the delivery vehicle is an unmanned vehicle.

Appendix 11

(311) The control device according to any one of appendices **1** to **10**, wherein when the delivery vehicle is located at a point different from the determined point in at least one of a period from a timing when the control device is started to a timing when the delivery vehicle is first allowed to accept the movement destination or a period from a timing when delivery to the delivery destination by the delivery vehicle is completed to the timing when the delivery vehicle is first allowed to accept the movement destination, the controller performs a control for moving the delivery vehicle toward the point, and the controller does not perform the control for moving the delivery vehicle toward the point during a period from the timing when the delivery vehicle first accepts the movement destination to the timing when the delivery to the delivery destination by the delivery vehicle is completed.

Appendix 12

(312) The control device according to appendix 11, wherein, when a delivery completion report notifying that the delivery to the delivery destination by the delivery vehicle is completed is obtained or when a predetermined time elapses from acquisition of an arrival report notifying that the delivery vehicle is arrived at the delivery destination, the controller determines that the delivery to the delivery destination by the delivery vehicle is completed.

Appendix 13

(313) The control device according to any one of appendices **1** to **12**, wherein when the control device is started, the controller determines that the state of the delivery vehicle is the state of not accepting the movement destination, when the shipping request is accepted, the controller allows the delivery vehicle to accept the movement destination and determines that the state of the delivery vehicle is changed to the state of accepting the movement destination, and when the arrival report notifying that the delivery vehicle is arrived at the delivery destination is received, the controller determines that the state of the delivery vehicle is changed to the state of not accepting the movement destination.

Appendix 14

(314) A control device for controlling a delivery vehicle, the control device including: a communicator that transmits, to a terminal device at a shipping location, a notification notifying that a start timing of preparation has arrived when the delivery vehicle arrives at a boundary of a region determined (i) by using, as a reference, the shipping location accepted as a movement destination and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at the shipping location.

Appendix 15

(315) The control device according to appendix 14, further including: a controller that performs a control for delaying an arrival time of the delivery vehicle to the shipping location on the basis of a response time from transmission of the notification to reception of a response to the notification.

Appendix 16

(316) A system for controlling a delivery vehicle, the system including: a determiner that determines a point (i) by using each of a plurality of shipping locations as a reference and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at each of the plurality of shipping locations; and a controller that allows the delivery vehicle to accept a shipping location of a package or a delivery destination of the package

as a movement destination so as to set a state of the delivery vehicle to a state of accepting the movement destination, moves the delivery vehicle to the movement destination when the state of the delivery vehicle is the state of accepting the movement destination, and moves the delivery vehicle toward the point when the state of the delivery vehicle is a state of not accepting the movement destination.

Appendix 17

(317) A system for controlling a delivery vehicle, the system including: a communicator that transmits, to a terminal device at a shipping location, a notification notifying that a start timing of preparation has arrived when the delivery vehicle arrives at a boundary of a region determined (i) by using, as a reference, the shipping location accepted as a movement destination and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at the shipping location.

Appendix 18

(318) A method to be performed by a control device or a system for controlling a delivery vehicle, the method including: a determination step of determining a point (i) by using each of a plurality of shipping locations as a reference and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at each of the plurality of shipping locations; and a control step of allowing the delivery vehicle to accept a shipping location of a package or a delivery destination of the package as a movement destination so as to set a state of the delivery vehicle to a state of accepting the movement destination, moving the delivery vehicle to the movement destination when the state of the delivery vehicle is the state of accepting the movement destination, and moving the delivery vehicle toward the point when the state of the delivery vehicle is a state of not accepting the movement destination.

Appendix 19

(319) A method to be performed by a control device or a system for controlling a delivery vehicle, the method including: a communication step of transmitting, to a terminal device at a shipping location, a notification notifying that a start timing of preparation has arrived when the delivery vehicle arrives at a boundary of a region determined (i) by using, as a reference, the shipping location accepted as a movement destination and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at the shipping location.

Appendix 20

(320) A control device for controlling a delivery vehicle, the control device including: a determiner that determines a point (i) by using each of a plurality of shipping locations as a reference and (ii) on the basis of a preparation time from when a shipping request is accepted until when shipment becomes possible at each of the plurality of shipping locations; and a controller that prevents execution of a control for moving the delivery vehicle toward the point while at least one of a collection task or a delivery task exists, the collection task being a task of allowing the delivery vehicle to collect a product, and the delivery task being a task of allowing the delivery vehicle to deliver the product, and performs the control for moving the delivery vehicle toward the point when the delivery vehicle is located at a point different from the determined point while the collection task and the delivery task do not exist.

Appendix 21

(321) The control device according to appendix 20, wherein when an order is accepted, the controller generates the collection task of allowing the delivery vehicle to collect the product at a shipping location specified in the order among the plurality of shipping locations and the delivery task of allowing the delivery vehicle to deliver the collected product to a delivery destination specified in the order, the control device further includes a communicator that receives a collection completion report notifying that collection of the product is completed and a delivery completion report notifying that delivery of the product is completed, and the controller deletes the collection task when the collection completion report is received and deletes the delivery task when the

delivery completion report is received.

REFERENCE SIGNS LIST

(322) **1** Delivery system **100**, **690**, **790** Control device **101**, **691** CPU **102**, **692** RAM **103***a*, **693***a* ROM **103***b* Hard disk **104**, **694** Data communication circuit **105***a*, **695***a* Video card **105***b*, **695***b* Display device **105***c*, **695***c* Input device **110** Acquirer **120** First Determiner **130** Controller **140** Second Determiner **190** Information storage **600**, **700** Delivery vehicle **601**, **602** Wheel **610** Chassis **620** Storage cabinet **621** Storage box **621***a* Door **621***b* Door frame **621***c* Dead bolt **621***d* Strike **630**, **730** Imaging device **641**, **741** LiDAR sensor **693***b* Flash memory **696** Position measurement circuit **698** Input/output port **699** Driving circuit **701** to **704** Propeller arm **711** to **714** Propeller **721***a* First holding frame **721***b* Second holding frame **722***a*, **722***b* Guide rail **800**, **901**, **902** Terminal device B**1**, B**1**′, B**2** Boundary D**1**, D**2**, D**2**′ Distance IN Internet L**1**, L**2** Region distance N**1**, N**1**′, N**2** Neighboring region PD Point R**1**, R**2** Shipping location ST**1**, ST**2** Store T**1**, T**2** Region time

Claims

1. A control device configured to control an unmanned delivery vehicle, the control device comprising: at least one memory storing program code; at least one processor configured to access the program code and operate as instructed by the program code; and a communication circuit configured to communicate with the unmanned delivery vehicle, wherein the program code includes: acquisition code configured to cause the at least one processor to acquire, for k=1 to K where k is a first integer and K is a second integer of 2 or more, k-th preparation time information representing a length of a k-th preparation time that is based on an assumption that a shipping request for an estimated product being a product of which shipment is estimated to be requested has been accepted at a k-th shipping location and that is a time from when the shipping request is accepted until when shipment of the estimated product becomes possible at the k-th shipping location, the k-th preparation time information being acquired from a storage storing the k-th preparation time information in advance; determination code configured to cause the at least one processor to: for k=1 to the K, determine a k-th boundary that is separated from the k-th shipping location by a distance in which the unmanned delivery vehicle moves in a duration represented by the acquired k-th preparation time information, and determine, as a point of an objective destination, a first point where a first sum of k-th distances for k=1 to the K that are each a k-th distance to the k-th boundary, or a second sum of k-th movement times for k=1 to the K that are each a k-th movement time of the unmanned delivery vehicle to the k-th boundary, decreases, in preference to a second point where the first sum or the second sum increases; and control code configured to cause the at least one processor to repeatedly perform a control for: based on a shipment request of a shipping product requested to be shipped is being accepted at a shipping location among first to K-th shipping locations, moving the unmanned delivery vehicle to the shipping location, based on a collection completion report notifying that collection of the shipping product is completed is-being received by the communication circuit, moving the unmanned delivery vehicle storing the shipping product to a delivery destination of the shipping product, and based on a delivery completion report notifying that delivery to the delivery destination is completed being received or based on a predetermined time elapsing from acquisition of an arrival report notifying that the unmanned delivery vehicle is arrived at the delivery destination, moving the unmanned delivery vehicle toward the point of the objective destination.

2. The control device according to claim 1, wherein the determination code is configured to cause the at least one processor to determine, as the point of the objective destination, a third point where a third sum of weighted k-th distances for k=1 to the K that are each the k-th distance weighted by a k-th probability that the shipping request is accepted for the k-th shipping location or a fourth sum of weighted k-th movement times for k=1 to the K that are each the k-th movement time

weighted by the k-th probability decreases, in preference to a fourth point where the third sum or the fourth sum increases.

3. The control device according to claim 2, wherein the determination code is configured to cause the at least one processor to determine the point of the objective destination such that the third sum or the fourth sum is minimized.

4. The control device according to claim 1, wherein the determination code is configured to cause the at least one processor to correct, for k=1 to the K, the length represented by the k-th preparation time information based on at least one of a number of employees on duty at the k-th shipping location or a number of packages for each of which the shipping request is accepted but preparation is not completed.

5. The control device according to claim 1, wherein when the K is 2, the storage: stores, in advance, first preparation time information representing a length of a first preparation time that is based on an assumption that the shipping request for the estimated product being a product of which shipment is estimated to be requested has been accepted at a first shipping location and that is a time from when the shipping request is accepted until when shipment of the estimated product becomes possible at the first shipping location, and stores, in advance, second preparation time information representing a length of a second preparation time that is based on an assumption that the shipping request for the estimated product being a product of which shipment is estimated to be requested has been accepted at a second shipping location and that is a time from when the shipping request is accepted until when shipment of the estimated product becomes possible at the second shipping location, the acquisition code is configured to cause, when the K is 2, the at least one processor to acquire the first preparation time information and the second preparation time information from the storage, and the determination code is configured to cause, when the K is 2, the at least one processor to: determine a first boundary that is separated from the first shipping location by a distance in which the unmanned delivery vehicle moves in a duration represented by the acquired first preparation time information, and determine a second boundary that is separated from the second shipping location by a distance in which the unmanned delivery vehicle moves in a duration represented by the acquired second preparation time information; determine, as a first region, the first boundary and a region closer to the first shipping location than the first boundary, and determine, as a second region, the second boundary and a region closer to the second shipping location than the second boundary; and in a first case, a second case, or a third case, determine, as the point of the objective destination, a third point where a third sum of a first distance being a distance to the first boundary and a second distance being a distance to the second boundary or a fourth sum of a first movement time being a movement time of the unmanned delivery vehicle to the first boundary, and a second movement time being a movement time of the unmanned delivery vehicle to the second boundary, decreases, in preference to a fourth point where the third sum or the fourth sum increases, wherein the first case is a case where the first region wholly covers the second region and a point of tangency does not exist between the first boundary corresponding to the first region and the second boundary corresponding to the second region, the second case is a case where the first region wholly covers the second region and the point of tangency does exist between the first boundary and the second boundary, and the third case is a case where the first region partially covers the second region but does not wholly cover the second region.

6. The control device according to claim 5, wherein the determination code is configured to cause, when the K is 2, the at least one processor to: in the first case where the first region wholly covers the second region and the point of tangency does not exist, determine the point of the objective destination at a first position located inside the first region and located outside the second region; in the second case where the first region wholly covers the second region and the point of tangency does exist, determine the point of the objective destination at a second position of the point of tangency; and in the third case where the first region partially covers the second region but does not wholly cover the second region, determine the point of the objective destination at a third position

of a point of intersection of the first boundary and the second boundary.
7. The control device according to claim 1, wherein the determination code is configured to cause the at least one processor to determine the point of the objective destination such that the first sum or the second sum is minimized.
8. The control device according to claim 1, wherein for k=1 to the K, the k-th preparation time at the k-th shipping location includes a search time required for finding at the k-th shipping location the estimated product of which shipment is estimated to be requested at the k-th shipping location, and a transportation time required for transporting the estimated product, includes the search time, a packaging time required for packaging the estimated product at the k-th shipping location, and the transportation time, or includes a cooking time required for cooking the estimated product at the k-th shipping location, the packaging time, and the transportation time.
9. The control device according to claim 1, wherein the control code is configured to cause the at least one processor to: allow the unmanned delivery vehicle to accept the shipping location among the first to K-th shipping locations or the delivery destination of the shipping product as a movement destination so as to set a state of the unmanned delivery vehicle to a first state of accepting the movement destination; move the unmanned delivery vehicle to the movement destination when the state of the unmanned delivery vehicle is the first state; and move the unmanned delivery vehicle toward the point of the objective destination when the state of the unmanned delivery vehicle is a second state of not accepting the movement destination.
10. The control device according to claim 9, wherein the control code is configured to cause the at least one processor to cause the unmanned delivery vehicle to move toward the point of the objective destination when the unmanned delivery vehicle is located at a third point different from the point of the objective destination in at least one of a first period from a first timing when the control device is started to a second timing when the unmanned delivery vehicle is first allowed to accept the movement destination or a second period from a third timing when delivery to the delivery destination by the unmanned delivery vehicle is completed to the second timing, and the control code is configured to cause the at least one processor not to cause the unmanned delivery vehicle to move toward the point of the objective destination during a third period from the second timing to the third timing.
11. The control device according to claim 10, wherein the control code is configured to cause the at least one processor to determine that the delivery to the delivery destination by the unmanned delivery vehicle is completed when the delivery completion report notifying that the delivery to the delivery destination by the unmanned delivery vehicle is completed is obtained or when a predetermined time elapses from acquisition of the arrival report notifying that the unmanned delivery vehicle has arrived at the delivery destination.
12. The control device according to claim 9, wherein the control code is configured to cause the at least one processor to, when the control device is started, determine that the state of the unmanned delivery vehicle is the second state, the control code is configured to cause the at least one processor to, when the shipping request is accepted, allow the unmanned delivery vehicle to accept the movement destination and determine that the state of the unmanned delivery vehicle is changed to the first state, and the control code is configured to cause the at least one processor to, when the arrival report notifying that the unmanned delivery vehicle has arrived at the delivery destination is received, determine that the state of the unmanned delivery vehicle is changed to the second state.
13. The control device according to claim 9, wherein the control code is configured to cause the at least one processor to, when the shipping request is accepted at the shipping location among the first to K-th shipping locations, allow the unmanned delivery vehicle to accept, as the movement destination, the shipping location at which the shipping request is accepted, and move the unmanned delivery vehicle to the movement destination.
14. A method to be performed by a control device or a system that includes at least one processor and a communication circuit configured to communicate with an unmanned delivery vehicle and is

for controlling the unmanned delivery vehicle, the method comprising: acquiring, for k=1 to K where k is a first integer and K is a second integer of 2 or more, k-th preparation time information representing a length of a k-th preparation time that is based on an assumption that a shipping request for an estimated product being a product of which shipment is estimated to be requested has been accepted at a k-th shipping location and that is a time from when the shipping request is accepted until when shipment of the estimated product becomes possible at the k-th shipping location, the k-th preparation time information being acquired from a storage storing the k-th preparation time information in advance; determining, for k=1 to the K, a k-th boundary that is separated from the k-th shipping location by a distance in which the unmanned delivery vehicle moves in a time of the length duration represented by the acquired k-th preparation time information; determining, as a point of an objective destination, a first point where a first sum of k-th distances for k=1 to the K that are each a k-th distance to the k-th boundary or a second sum of k-th movement times for k=1 to the K that are each a k-th movement time of the unmanned delivery vehicle to the k-th boundary, decreases, in preference to a second point where the first sum or the second sum increases; and repeatedly performing a control for: based on a shipment request of a shipping product requested to be shipped is being accepted at a shipping location among first to K-th shipping locations, moving the unmanned delivery vehicle to the shipping location, based on a collection completion report notifying that collection of the shipping product is completed being received by the communication circuit, moving the unmanned delivery vehicle storing the shipping product to a delivery destination of the shipping product, and based on a delivery completion report notifying that delivery to the delivery destination is completed being received or based on a predetermined time elapsing from acquisition of an arrival report notifying that the unmanned delivery vehicle is arrived at the delivery destination, moving the unmanned delivery vehicle toward the point of the objective destination.

15. The method according to claim 14, further comprising: allowing the unmanned delivery vehicle to accept the shipping location among the first to K-th shipping locations or the delivery destination of the shipping product as a movement destination so as to set a state of the unmanned delivery vehicle to a first state of accepting the movement destination; moving the unmanned delivery vehicle to the movement destination when the state of the unmanned delivery vehicle is the first state; and moving the unmanned delivery vehicle toward the point of the objective destination when the state of the unmanned delivery vehicle is a second state of not accepting the movement destination.